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Evidence from ESG-Screened Loan Portfolios

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Abstract

This paper asks whether the ESG performance of European banks is associated with lower credit risk in their lending portfolios. We estimate two-way fixed-effects regressions of the non-performing loan (NPL) ratio on (i) Bloomberg ESG scores for 35 listed European commercial banks over 2015–2024 (Model 1), and (ii) the Rated ESG Exposure Index (REEI), a novel composite that weights each bank’s sectoral loan exposure by climate transition risk, hand-collected from Pillar 3 disclosures for 32 banks over 2018–2024 (Model 2). In Model 1, the combined ESG coefficient is negative across all specifications and reaches significance under the post-2018 IFRS 9 subsample and the horse-race overlap-sample specifications. Lead-ESG and parallel lead-S placebos indicate that rating methodologies absorb prior credit-stress signals at both the combined-score and Social-pillar levels; we therefore interpret all Model 1 results, including the Social-pillar finding, as associations rather than causal effects and treat the contemporaneous specification as the specification most exposed to this channel. The pillar decomposition is the most informative result: the Social pillar drives a strong negative association (-1.19 , $p < 0.001$) that survives the lead-S placebo with a much smaller discount than the combined score ($\approx 1.3\times$ vs $\approx 4.6\times$), while the Environmental pillar enters positive, a pattern we attribute to the disclosure-weighted construction of the Bloomberg E score rather than to environmental management raising credit risk. Model 2 produces a clean null: the REEI coefficient is statistically indistinguishable from zero and reverses sign on the exclusion of four distressed banks. A horse-race specification on the 149-observation overlap sample shows that the two regressors capture largely independent within-bank variation (correlation 0.27); joint inclusion barely affects the ESG coefficient. The paper contributes by empirically separating the market-perception channel (ESG ratings) from the portfolio-composition channel (sectoral transition exposure), and by documenting that the credit-risk-relevant content of the aggregate ESG score is concentrated in the Social pillar rather than the Environmental pillar.

Keywords: ESG scores, credit risk, non-performing loans, European banks, climate transition risk, Pillar 3 disclosures, panel data, fixed effects.

JEL Classification: G21, G32, Q56.

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List of Abbreviations

AML	Anti-Money Laundering
BECS	Bloomberg ESG Classification System
BIS	Bank for International Settlements
CDS	Credit Default Swap
CET1	Common Equity Tier 1
CQ5	Pillar 3 ESG template: sectoral exposures by NACE code
CR1-B	Pillar 3 template: credit risk by exposure class (pre-CQ5)
CRR	Capital Requirements Regulation
CRW	Climate Risk Weight
EBA	European Banking Authority
ECB	European Central Bank
ESG	Environmental, Social, and Governance
EU	European Union
FE	Fixed Effects
GDP	Gross Domestic Product
GHG	Greenhouse Gas
GMM	Generalised Method of Moments
HC1	Heteroskedasticity-Consistent (Huber–White type 1) standard errors
HICP	Harmonised Index of Consumer Prices
IAS 39	International Accounting Standard 39 (Financial Instruments)
IFRS 9	International Financial Reporting Standard 9 (Financial Instruments)
ITS	Implementing Technical Standards
LogTA	Natural Logarithm of Total Assets
NACE	Statistical Classification of Economic Activities in the European Community
NIM	Net Interest Margin
NPE	Non-Performing Exposure
NPL	Non-Performing Loan
OECD	Organisation for Economic Co-operation and Development
ONS	Office for National Statistics (UK)
REEI	Rated ESG Exposure Index
ROA	Return on Assets
ROE	Return on Equity
SASB	Sustainability Accounting Standards Board
SE	Standard Error
VIF	Variance Inflation Factor

1. Introduction

The relationship between environmental, social, and governance (ESG) factors and financial performance has become one of the most actively researched topics in corporate finance over the past decade. While much of this literature focuses on equity markets, examining whether ESG performance predicts stock returns, reduces volatility, or lowers the cost of equity capital, the credit risk dimension has received comparatively less attention. This gap is surprising given the central role that banks play in the European financial system and the growing regulatory pressure on banks to integrate sustainability considerations into their risk management frameworks.

The urgency of this question stems from two converging developments. First, the European regulatory landscape for sustainable finance has evolved rapidly. The EU Green Deal, the ECB’s Guide on Climate-Related and Environmental Risks (2020), and the EBA’s binding standards for Pillar 3 ESG disclosures (ITS on ESG disclosures, effective from 2023 for large institutions) collectively signal that climate-related financial risk is no longer a peripheral concern but a core element of prudential supervision. The ECB’s economy-wide climate stress test in 2022, which assessed the vulnerability of euro area banks to physical and transition risks, further underscored the systemic relevance of these issues. Second, the concept of “green swan” events, climate-related disruptions that are foreseeable in nature but deeply uncertain in timing, magnitude, and interaction effects, has entered the central banking discourse through Bolton et al. (2020). Unlike traditional black swans, green swans derive their systemic danger not from unpredictability alone but from the interaction between physical risks, transition risks, and the financial system’s exposures to climate-policy-relevant sectors.

Against this backdrop, the central research question of this paper is: *Does the ESG performance of European banks predict lower credit risk in their lending portfolios?* We operationalise credit risk as the non-performing loan (NPL) ratio, the share of non-performing loans in a bank’s gross lending portfolio, which captures realised credit deterioration rather than market-implied default probabilities. While NPL ratios are backward-looking and subject to accounting regime changes (most notably the transition from IAS 39 to IFRS 9 in 2018), they remain the most widely available and comparable measure of loan portfolio quality across European banks.

Our empirical approach employs a two-model structure that separates two distinct channels through which ESG may relate to credit risk. Model 1, our main specification, tests the association between Bloomberg ESG pillar scores and the NPL ratio across 35 listed European commercial banks over 2015–2024, using two-way fixed effects panel regressions with bank and year fixed effects. This model captures the “market perception” channel: it asks whether banks that are perceived as more sustainable by ESG rating agencies also exhibit lower credit risk. The ESG score is a summary statistic produced by an external rater (Bloomberg), and any observed association could reflect either genuine risk management practices, signalling effects, or common drivers that affect both ESG scores and loan quality.

Model 2 complements this with a “balance-sheet mechanism” test. Using hand-collected Pillar 3 disclosures from a subsample of European banks, we construct the Rated ESG Exposure Index (REEI), a composite measure that weights each bank’s sectoral loan exposure (reported at the NACE sector level) by climate transition risk weights (CRW). The CRW calibration draws on the sector reclassification framework introduced by Battiston et al. (2017), who mapped NACE

Rev2 economic activities into climate-policy-relevant categories (fossil fuels, utilities, energy-intensive industries, transport, and housing). A bank with a high REEI has a lending portfolio concentrated in sectors that face elevated transition risk under decarbonisation scenarios; a bank with a low REEI lends predominantly to sectors with limited climate policy exposure. This model asks whether the composition of a bank’s lending portfolio, independent of its ESG label, predicts credit deterioration.

The two-model structure represents the paper’s primary methodological contribution. While prior studies have examined the ESG–credit risk link using either firm-level ESG ratings (Hauch, 2026; Apergis et al., 2022) or portfolio-level climate exposure metrics (Battiston et al., 2017), few have combined both approaches within a single study. By testing perception and mechanism channels separately, we can begin to disentangle whether ESG ratings capture genuine risk-relevant information about lending portfolios or whether they primarily reflect reputation, disclosure quality, or other bank characteristics that are correlated with but conceptually distinct from underlying credit risk.

Results from Model 1, based on the full sample (2015–2024, 347 bank-year observations for the contemporaneous specification and 314 observations for the lagged specification), indicate that the coefficient on the combined ESG score is negative across all four nested specifications, with point estimates ranging from -0.62 to -0.31 . A one-standard-deviation increase in the lagged Bloomberg ESG score (1.19 points on the Bloomberg 0–10 scale) is associated with a decrease in the NPL ratio of approximately 0.44 percentage points in the saturated specification. However, the estimates do not reach conventional levels of statistical significance (p -values between 0.28 and 0.57). This finding, a directionally consistent but statistically insignificant association, is itself informative. With a cross-section of only 35 banks, the study has limited statistical power to detect effects that are economically meaningful but modest in magnitude. For comparison, Hauch (2026), who found statistically significant effects of ESG divergence on CDS spreads, worked with a panel of 562 firms. The power difference is substantial: detecting a coefficient of similar magnitude with 35 entities requires an effect size roughly four times larger to achieve the same statistical significance. The pillar decomposition (Section 4) reveals additional heterogeneity: the Social pillar is strongly negative and statistically significant, the Governance pillar is negative but insignificant, and the Environmental pillar is positive and statistically significant. The interpretation of this pillar pattern, which we discuss in detail in Section 4, requires careful consideration of the architecture of Bloomberg’s ESG performance score and is one motivation for our complementary Model 2 (REEI) analysis. We further note that a lead-ESG placebo specification (regressing $NPL_{i,t}$ on $ESG_{i,t+1}$) yields a large and significant negative coefficient, indicating that rating-agency methodologies incorporate financial-stress signals into subsequent assessments; a parallel lead-S placebo on the Social pillar confirms that the same dynamic operates within S, though the lead-S/lagged-S magnitude ratio (≈ 1.3) is much smaller than the lead-ESG/lagged-ESG ratio (≈ 4.6), so a substantial forward-looking component of the Social pillar survives the test. We therefore interpret all Model 1 estimates throughout the paper as associations rather than as causal estimates of an ESG-to-NPL channel.

Model 2, estimated on a hand-collected Pillar 3 panel of 151 bank-year observations across 32 banks over 2018–2024, produces REEI coefficients that are negative in point-estimate terms

but never reach significance after controls; the negative point estimate flips sign on the exclusion of four distressed banks (Danske Bank, Swedbank, Monte dei Paschi di Siena, Eurobank), confirming that the result is fragile and best characterised as a clean null. We are cautious about reading this null as evidence against the underlying transition-risk mechanism, since the result reflects the joint limitations of a short and unbalanced seven-year window, a small cross-section, and the slow-moving nature of the REEI; a reading consistent with these limitations is that the realised credit consequences of an ESG-misaligned loan book have not yet materialised in NPL ratios over our window, in line with the long-horizon framing of climate transition risk. The most empirically interesting result emerges from a horse-race specification on the identical 149-observation overlap sample where both models can be jointly estimated: the lagged ESG score is significant (-1.00 , $p = 0.043$), the REEI is not, and joint inclusion of the two regressors barely affects the ESG coefficient. The within-bank correlation between the two regressors is only 0.27, indicating that, on our data, the empirical variation generated by ESG ratings and by sectoral transition-risk exposure is largely non-overlapping: the perception channel and the mechanism channel are not the same channel re-measured, though the conceptual distinction does not require them to be fully independent in every setting.

Beyond its two-model structure, the paper makes three further contributions to the literature. First, by focusing on banks rather than non-financial corporates, the study extends an ESG–credit risk literature that has been dominated by firm-level studies (Hauch, 2026; Apergis et al., 2022) into a setting (bank lending) where the link operates differently because banks are simultaneously the rated entity and the intermediary whose lending decisions shape the ESG profile of the broader economy. Second, the hand-collected Pillar 3 data enable a granular sectoral analysis that is not possible with standard financial databases, which typically report only aggregate loan volumes; the REEI itself is a methodological contribution that, independent of its empirical performance in our short panel, can be applied as a supervisory and analytical instrument as Pillar 3 ESG disclosures generate longer standardised time series. Third, the pillar decomposition documents that, in our European bank panel, the credit-risk-relevant content of the Bloomberg aggregate ESG score is concentrated in the Social pillar rather than the Environmental pillar, qualifying earlier readings of the ESG–credit-risk link that treat the aggregate score as a homogeneous risk signal (Liu & Xie, 2024; Al-Qudah et al., 2022) and complementing the rating-divergence and measurement-error concerns documented by Berg et al. (2022, 2024) and Kotsantonis & Serafeim (2019). The study also explicitly addresses the IFRS 9 structural break in NPL recognition, which complicates any panel study spanning the 2018 transition year, through both year fixed effects (which absorb the level shift) and a dedicated post-2018 subsample robustness check.

The remainder of the paper is structured as follows. Section 2 reviews the relevant literature on ESG and financial performance, ESG and credit risk, ESG rating divergence, climate transition risk, and the role of social norms in financial markets. Section 3 describes the methodology, including the sample construction, variable definitions, REEI construction process, and econometric specification. Section 4 presents the results from both models and the robustness checks. Section 5 discusses the findings in the context of the existing literature and draws out policy implications. Section 6 concludes.

2. Literature Review

2.1 ESG and Financial Performance: The Broader Landscape

The relationship between ESG performance and financial outcomes has been examined extensively in the academic literature, yet there is still no clear consensus on whether ESG improves financial performance. Meta-analyses such as Friede et al. (2015) show that most studies find a non-negative relationship between ESG criteria and corporate financial performance. However, the magnitude and significance of the effect vary considerably depending on the ESG measure, the financial outcome considered, the sample period, and the geographical setting.

Several theoretical explanations have been proposed for why ESG performance may affect financial outcomes. Stakeholder theory (Freeman, 1984) argues that firms that manage relationships with employees, customers, communities, and the environment effectively are better able to reduce operational and reputational risks. Over time, this can lower financing costs and improve profitability. The risk mitigation view suggests that strong ESG practices serve as a form of insurance against tail risk. Strong ESG performance may help firms avoid costly events such as environmental liabilities, regulatory penalties, or labour disputes that could otherwise damage firm value.

At the same time, a competing view rooted in neoclassical finance argues that ESG investments may represent agency costs. From this perspective, managers allocate resources to socially desirable but financially inefficient activities, ultimately reducing shareholder value (Friedman, 1970). Under this interpretation, positive ESG-performance relationships observed in empirical studies may reflect reverse causality, since more profitable firms have greater resources to invest in ESG initiatives, or omitted variable bias, where both ESG performance and financial outcomes are driven by factors such as management quality. Concerns related to endogeneity are therefore central to the ESG literature and are also relevant for this study, as discussed in Section 3.

Within the banking sector, the literature on ESG and financial performance is comparatively limited. Banks differ from non-financial firms in several important ways that complicate the analysis. They operate in a heavily regulated environment, their balance sheets are relatively opaque, and their core business consists of intermediating credit risk. In addition, a bank's ESG impact is determined less by its own operations and more by the activities it finances through lending and investment portfolios. For example, a bank's environmental footprint depends largely on the sectors to which it extends credit. This distinction is particularly important for our study and motivates the construction of the REEI index in Model 2. Existing studies on ESG and bank performance mainly focus on profitability measures such as ROA and ROE or on market valuation measures such as Tobin's Q, while realised credit risk measures such as non-performing loan ratios remain comparatively underexplored.

Institutional evidence further underlines the growing importance of climate-related risks for the banking sector. In 2022, the European Central Bank conducted a comprehensive climate risk stress test to assess banks' exposure and preparedness with respect to climate risks (ECB, 2022). The results show that climate-related risks are highly relevant for European banks because a substantial share of their activities is connected to carbon-intensive sectors. On average, more than 60% of banks' corporate lending income is linked to greenhouse-gas-emitting industries,

implying considerable exposure to transition risk.

The stress test also demonstrates that banks are vulnerable to both transition risks and physical climate risks, which can affect financial stability through credit risk channels. Under adverse climate scenarios, the ECB estimates that combined credit and market losses could amount to approximately €70 billion, although this figure likely understates the true level of risk because of data and modelling limitations. The report further concludes that many banks are still at an early stage of incorporating climate risk into their internal credit risk models. In many cases, risk parameters remain insufficiently sensitive to climate-related shocks.

These findings highlight significant gaps in current risk measurement practices and reinforce the need for more refined approaches to assessing climate-related credit risk exposure. They also provide strong institutional support for the idea that climate-related factors, including ESG performance and sectoral transition exposure, are becoming increasingly important determinants of bank credit risk. This motivates empirical research that links ESG indicators and sector-based transition risk measures, such as the REEI introduced in Section 3.5, to observed credit risk outcomes in European banks.

2.2 ESG and Credit Risk

The literature linking ESG performance to credit risk is relatively recent and particularly relevant for our research question. Two studies are especially important from a methodological perspective and serve as key references for our empirical design.

Hauch (2026) analyses the relationship between ESG rating divergence and corporate credit risk using a panel of 562 publicly listed non-financial firms between 2017 and 2022. The study applies fixed-effects regressions with credit default swap (CDS) spreads and probability of default as dependent variables. ESG rating divergence, measured as the pairwise standard deviation across ESG ratings from Refinitiv, MSCI, S&P, and Bloomberg, is used as the key explanatory variable.

The results show that higher ESG rating divergence is associated with higher credit risk. A one-standard-deviation increase in divergence leads to an increase of approximately 0.8 to 1.3 basis points in five-year CDS spreads. At the ESG pillar level, the effect is mainly driven by the environmental dimension. Divergence in environmental scores increases CDS spreads by roughly 1.5 basis points, while divergence in the social and governance dimensions does not produce statistically significant effects. The study also finds that credit ratings only partially mediate the relationship between ESG divergence and credit risk, suggesting that the effect operates largely through direct market pricing. In addition, stronger country-level ESG performance reduces the risk premium associated with ESG divergence by around 50–65%.

Hauch (2026) represents the closest methodological reference for our Model 1. Several elements of our empirical setup are based on this framework, including the use of fixed effects to control for time-invariant firm or bank characteristics, the inclusion of both aggregate and pillar-level ESG measures, and the use of lagged ESG variables to mitigate reverse causality concerns. At the same time, our study differs in several important respects. First, we focus on banks rather than non-financial firms, meaning that our dependent variable captures realised credit risk in the bank’s lending portfolio instead of market-implied default risk. Second, our sample

is considerably smaller, consisting of 35 banks rather than more than 500 firms, which affects statistical power. Third, our analysis extends beyond ESG rating analysis by incorporating the REEI-based Model 2, which focuses on sectoral transition risk exposure.

Apergis et al. (2022) examine the relationship between ESG scores and the cost of debt for S&P 500 firms over the period 2010–2019. Using Refinitiv ESG pillar scores and bond spreads, the authors estimate restricted maximum likelihood mixed-effects regressions that account for both fixed and random effects. Their results show that all three ESG pillars are negatively and significantly associated with bond yields. The estimated coefficients for the environmental, social, governance, and combined ESG scores are approximately -0.63 , -0.94 , -0.77 , and -1.16 , respectively.

Despite these findings, traditional credit measures remain substantially more important in explaining bond spreads. Credit ratings have coefficients roughly nine times larger than those associated with ESG variables, suggesting that ESG information contributes only marginally to explaining credit risk relative to conventional financial indicators. For our study, Apergis et al. (2022) provide two important benchmarks. First, the estimated ESG coefficients offer a useful point of comparison for interpreting our own results. Second, the finding that ESG explains only a limited share of credit risk variation suggests that ESG should be viewed as a supplementary rather than dominant risk signal. This is particularly relevant in the context of our smaller sample of European banks, where within-bank ESG variation over time may be relatively limited.

A recent systematic review by Carnevale & Drago (2024) provides a broader overview of whether banks incorporate ESG risks into lending decisions. The authors conclude that banks generally penalise poor ESG performance through higher loan spreads, indicating that ESG risks are at least partly priced in credit markets. At the same time, the evidence remains mixed. Strong ESG performance does not consistently reduce borrowing costs, and empirical findings vary considerably across institutional settings and study designs.

The review also highlights several important limitations in the existing literature. Most studies focus on syndicated loans to large listed firms and rely heavily on loan pricing measures such as spreads and fees. By contrast, realised credit risk indicators such as non-performing loan ratios receive comparatively little attention. These limitations are directly relevant for our study and support our decision to focus on realised credit risk and sector-level transition exposure.

A more direct connection between sustainable lending and realised credit risk is provided by Al-Qudah et al. (2022), who analyse the effect of green lending on non-performing loans in UAE banks between 2015 and 2020. Using a sample of 23 banks and estimating both random-effects and two-stage least squares (2SLS) regression models, the authors find a statistically significant negative relationship between the share of green loans and NPL ratios. Banks with a larger proportion of green lending tend to exhibit lower credit risk.

This study is particularly relevant for our research design for several reasons. Unlike much of the ESG-credit risk literature, which relies on market-based indicators such as CDS spreads or bond yields, Al-Qudah et al. (2022) focus on realised credit outcomes (NPL ratios), aligning closely with our dependent variable. In addition, the study uses loan portfolio composition (green loan ratios) as its key explanatory variable, which is conceptually close to our REEI framework. Their findings suggest that sustainability-oriented lending may improve portfolio

quality by directing credit toward lower-risk or more resilient sectors.

However, the study also has limitations. The analysis is restricted to a single country (UAE) and uses a relatively small sample, which raises concerns about external validity. Furthermore, the distinction between “green” and “non-green” lending does not fully capture the continuous spectrum of transition risk exposure across sectors. Our REEI index is designed to address this limitation by extending the analysis to a broader European banking context and incorporating a more granular measure of sectoral climate risk; our study builds on and generalises the insights of Al-Qudah et al. (2022).

Additional evidence on the relationship between ESG and bank risk is provided by Liu & Xie (2024), who investigate how ESG performance affects liquidity risk in Chinese listed banks. The authors find that banks with higher ESG scores tend to exhibit lower liquidity risk, primarily because ESG performance improves asset quality and financial performance. In particular, stronger ESG performance is associated with lower non-performing loans and higher profitability, both of which improve banks’ resilience to liquidity shocks.

The study also sheds light on the mechanisms through which ESG affects bank risk. Banks with stronger ESG performance appear more likely to incorporate sustainability considerations into lending decisions, thereby reducing exposure to high-risk borrowers and mitigating problems related to adverse selection and moral hazard. At the same time, ESG engagement may strengthen stakeholder trust, stabilise funding sources such as customer deposits, and reduce funding volatility. These findings reinforce the expectation that ESG-oriented banks are likely to exhibit lower NPL ratios and support the relevance of our empirical framework.

Another important contribution is provided by Hoepner et al. (2016), who analyse the relationship between sustainability and the cost of debt using 470 loan agreements across 28 countries. Their findings reveal that country-level sustainability has a strong and statistically significant negative effect on borrowing costs, with a one-unit increase in sustainability scores leading to a reduction of approximately 64 basis points in loan spreads. In contrast, the authors find no robust evidence that firm-level sustainability significantly affects the cost of bank loans, highlighting the difficulty of capturing firm-specific sustainability risks using aggregated ESG measures.

This suggests that aggregate ESG measures at the firm level may not fully capture the dimensions of sustainability that are most relevant for credit risk. The authors argue that sustainability can influence credit markets through several channels, including lower information asymmetry, stronger trust between lenders and borrowers, and lower expected default risk. Overall, their findings point to important limitations in conventional ESG metrics and support the use of more targeted measures of transition-related exposure, such as the REEI index used in our study. This may be necessary to identify the true relationship between sustainability and credit risk.

2.3 ESG Rating Divergence and Data Quality

One of the main challenges in ESG research is the substantial divergence in ESG ratings across providers. Berg et al. (2022), in a widely cited study published in the *Review of Finance*, analyse the extent and sources of disagreement among six major ESG rating agencies: KLD, Sustainalytics, Moody’s ESG, S&P Global, Refinitiv, and MSCI. The authors decompose ESG rating divergence into three components: scope, measurement, and weight. Scope refers to the

specific ESG attributes included by each provider, measurement captures how the same attribute is assessed, and weight reflects differences in aggregation methods.

Their results show that most disagreement stems from measurement differences, which account for 56% of total divergence. Scope differences explain 38%, while weighting differences contribute only 6%. This means that even when rating agencies evaluate the same ESG category, such as carbon emissions or labour practices, they often arrive at substantially different conclusions. The average pairwise correlation between ESG ratings in their sample is 0.54, ranging from 0.38 to 0.71. For comparison, credit ratings from Moody’s and S&P typically correlate above 0.95.

Berg et al. (2022) further identify a “rater effect”: an agency’s overall assessment of a firm tends to influence its measurement of specific categories, akin to a halo effect in performance evaluation. This finding suggests that measurement divergence is not random noise but contains systematic, rater-specific biases. The governance dimension shows the lowest average correlation across rating agencies, with an average correlation of only 0.30. This is particularly important given the theoretical relevance of governance for credit risk and financial stability.

These findings have direct implications for our study. By relying on Bloomberg as a single ESG data provider, we ensure methodological consistency across all banks in the sample. However, this comes at the cost of reduced generalisability, since ESG scores may differ substantially across providers. Our results are conditional on Bloomberg’s specific scope, measurement and weighting choices. If alternative data sources such as Refinitiv, MSCI, or Sustainalytics had been used, the estimated relationships between ESG and credit risk might also differ in magnitude or statistical significance. A multi-provider approach similar to Hauch (2026) would likely improve robustness but was not feasible within the scope of this study. Further details on our choice of ESG data provider are discussed in Section 3.4.

Kotsantonis & Serafeim (2019) complement the academic analysis of Berg et al. (2022) by focusing on practical challenges related to ESG data quality. The authors identify four underappreciated issues. First, ESG coverage is uneven across firms and sectors, with smaller firms and firms from emerging markets often underrepresented. Second, ESG scores are largely backward-looking because they are based on historical disclosures rather than forward-looking risk assessments. Third, investors frequently confuse ESG risk exposure with ESG risk management. Finally, ESG scores are often influenced by firm size, as larger companies typically have greater resources for ESG reporting and disclosure.

Some of these concerns are less severe in our setting because the sample consists of large listed European banks that receive broad Bloomberg coverage. Nevertheless, the backward-looking nature of ESG scores remains highly relevant. For example, a bank’s ESG score in 2020 may largely reflect disclosures from 2019 or earlier, creating a timing mismatch with contemporaneous non-performing loan ratios. Our lagged ESG robustness check, by using ESG scores from period $t - 1$, partially addresses this concern.

More recent research has extended the discussion of ESG divergence by treating ESG ratings explicitly as a measurement-error problem. Berg et al. (2024) argue that ESG ratings should not be interpreted as precise measures of underlying ESG performance but rather as noisy proxies that contain substantial measurement error. Using ESG data from multiple providers, the

authors show that low agreement across ratings has important econometric implications.

In a classical errors-in-variables framework, regression models that use ESG scores as explanatory variables suffer from *attenuation bias*, meaning that estimated coefficients are biased toward zero. To address this issue, Berg et al. (2024) propose an instrumental-variable approach in which ESG ratings from alternative providers are used as instruments for one another. Their results indicate that corrected ESG coefficients are, on average, more than twice as large as conventional estimates.

These findings are highly relevant for our analysis. First, they suggest that the relationship between ESG and credit risk may be systematically underestimated in empirical studies, particularly in smaller samples such as ours. Second, our reliance on a single ESG provider increases the risk that the estimated coefficients are affected by measurement error, which limits the strength of statistical inference. Finally, the results imply that insignificant ESG coefficients should be interpreted cautiously, since they may reflect limitations in ESG measurement rather than the absence of a meaningful economic relationship.

2.4 Climate Transition Risk and the Financial System

The theoretical foundation for our Model 2 (the REEI-based test) draws heavily on two seminal contributions to the climate finance literature.

Battiston et al. (2017), published in *Nature Climate Change*, develop a network-based climate stress-testing framework for the financial system. A central contribution of the paper is the reclassification of NACE Rev. 2 economic activities into five sectors that are particularly relevant from a climate-policy perspective: fossil fuels, utilities, energy-intensive industries, transport, and housing. This classification is based on three criteria: direct greenhouse gas emissions (Eurostat Scope 1 data), the sector’s role within the energy supply chain, and its exposure to carbon leakage risk under the European Commission’s 2014 directive.

The authors show that although direct equity exposure of European financial institutions to the fossil fuel sector is relatively limited (4–13% of equity portfolios), combined exposure to climate-policy-relevant sectors is substantial (36–48%), with the energy-intensive sector accounting for the largest share. For banks, the primary transmission channel is lending rather than equity holdings. Battiston et al. (2017) estimate that loans to climate-policy-relevant sectors amount to a considerable share of Euro Area bank capital, including approximately 11.4% for fossil fuels and utilities, 28% for energy-intensive industries, 16% for transport, and 73% for housing.

The main methodological insight we draw from Battiston et al. (2017) is the idea that standard sector classifications do not automatically correspond to climate-risk categories. Certain activities classified within the same NACE sector can differ substantially in their exposure to transition risk. For example, some activities classified under B (Mining) are only weakly climate-relevant, whereas certain activities classified under C (Manufacturing), such as refined petroleum products, are highly exposed to transition risk.

Our REEI framework builds directly on this logic. Instead of using a simple binary classification of climate-relevant versus non-climate-relevant sectors, we assign Climate Risk Weights (CRWs) to NACE sectors on a continuous scale ranging from 0.1 to 1.0. This approach allows us

to capture varying degrees of transition risk exposure across sectors and provides a more nuanced representation of climate-related credit risk in bank loan portfolios.

Bolton et al. (2020), widely known as the “Green Swan” report published by the Bank for International Settlements, provide the broader conceptual foundation for why climate transition risk matters for financial stability and bank credit risk. The green swan metaphor captures climate-related financial risks that share some features of Taleb’s black swans (they can be highly disruptive) but differ in a crucial respect: they are not truly unpredictable. Climate change and the transition toward a lower-carbon economy have been discussed for decades, and the existence of transition risks is widely recognised. What remains uncertain is the timing, speed, and exact pathway of the transition, as well as the interaction between climate policy and financial market exposures.

Bolton et al. (2020) argue that traditional risk assessment methods, including Value-at-Risk models, conventional stress testing, and scenario analysis, face important limitations when applied to climate-related risks (“green swan risks”). These tools rely heavily on historical data and probability distributions, which may not adequately capture future outcomes characterised by non-linear climate effects, tipping points, and abrupt policy changes.

This perspective is important for our study because it supports our decision to interpret the relationship between ESG and non-performing loans primarily as an associative rather than strictly causal relationship. Given the uncertainty surrounding future climate transition pathways, establishing clear causal links is particularly challenging, even when advanced econometric methods are applied. The report also emphasises the importance of improved climate-related disclosure, especially detailed information on sectoral loan exposures. This is directly relevant for our study, as the REEI framework relies on sectoral exposure data extracted from banks’ Pillar 3 reports.

These arguments are further reinforced by the ECB’s 2022 climate risk stress test (ECB, 2022), which provides direct supervisory evidence on European banks’ exposure to climate-related risks. The ECB documents that banks not only derive a large share of their lending income from carbon-intensive sectors, but also remain insufficiently advanced in integrating climate risk into their internal risk models. This suggests that existing risk assessment frameworks may underestimate climate-related credit risk, particularly in portfolios with high sectoral concentration.

Taken together, the literature suggests that climate-related credit risk is transmitted primarily through sectoral exposure rather than through firm-level ESG characteristics alone. This provides the key rationale for using a sector-based measure such as the REEI index, which aims to capture transition risk exposure in bank lending portfolios in a more direct and economically meaningful way.

2.5 ESG in Insurance and Financial Value Chains

While a large part of the ESG literature focuses on firm-level ESG performance or investment portfolios, a growing body of research examines how sustainability considerations are incorporated into financial intermediation processes. Aburto Barrera & Wagner (2023) provide a systematic review of sustainability issues across the insurance and pension fund sectors and show

that ESG integration mainly takes place through risk management, investment decisions, and reporting practices.

Although the study focuses on insurance companies and pension funds rather than banks, many of the underlying mechanisms are directly relevant for the banking sector. ESG considerations influence how financial institutions evaluate risk exposures, allocate capital, and structure financial products. In the case of banks, the most important transmission channel is lending. ESG-related factors can affect borrower selection, sector allocation, and credit risk assessment, thereby influencing the overall risk profile of a bank’s loan portfolio.

This perspective supports the idea that the sustainability impact of banks is largely indirect and depends primarily on the activities they finance rather than on their own operational footprint. In other words, the composition of a bank’s lending portfolio may be more relevant for understanding sustainability-related risk than the bank’s internal ESG policies alone.

van Duuren et al. (2016) further analyse how ESG factors are integrated into investment and financial decision-making processes. Surveying conventional asset managers, the authors show that ESG integration has evolved beyond simple exclusion strategies toward approaches that incorporate ESG information directly into fundamental analysis and risk assessment frameworks; in their words, that ESG investing in practice closely resembles fundamental investing. This development is particularly relevant for banks, where ESG considerations are increasingly embedded in credit evaluation and lending decisions rather than treated as a separate screening exercise.

These insights are important for our study for two main reasons. First, ESG scores may capture not only a bank’s internal sustainability practices but also its ability to manage ESG-related risks across different business activities, including lending. Second, and more importantly, the literature highlights that sector allocation decisions are a key channel through which sustainability considerations affect financial risk. This directly motivates the use of the REEI index in our analysis. By focusing on the sectoral composition of bank loan portfolios, the REEI provides a more targeted measure of transition-related credit risk exposure than aggregate ESG scores alone.

2.6 Social Norms and Market Pricing of “Sin”

Hong & Kacperczyk (2009), in their influential study published in the *Journal of Financial Economics*, examine how social norms affect stock prices and returns of so-called “sin” industries, including alcohol, tobacco, and gaming companies. The authors show that these firms are less frequently held by norm-constrained investors such as pension funds and receive less analyst coverage than comparable firms. At the same time, sin stocks generate higher expected returns, which the authors interpret as a compensation premium for holding socially controversial assets. The estimated excess return is approximately 21 basis points per month, or around 2.5% annually, and the effect is observed not only in the United States but also in several international markets, including Canada, France, and Germany.

The relevance of Hong & Kacperczyk (2009) for our study lies in the broader idea that social norms can influence financial markets and the pricing of risk. Applied to the banking sector, this raises the question of whether banks may face similar consequences when their lending portfolios

are concentrated in sectors that are increasingly perceived as environmentally or socially harmful. Such effects could emerge through higher funding costs, stronger regulatory pressure, reputational concerns, or higher credit risk exposure.

This argument is particularly relevant for fossil fuel industries, which have increasingly become the focus of divestment campaigns, regulatory intervention, and public scrutiny. The fossil fuel sector, in particular, has undergone a transition in social perception analogous to tobacco in the 1960s: from a mainstream industry to one subject to divestment campaigns, regulatory tightening, and reputational pressure. Our REEI index, which assigns the highest Climate Risk Weight (1.0) to fossil fuels and mining, directly captures this dimension.

The comparison between “sin stocks” and “sin lending” is imperfect. Lending activities are generally less visible than equity investments, and climate transition risks are likely to materialise over a longer time horizon than the regulatory changes historically affecting tobacco or gambling industries. Nevertheless, the underlying mechanism remains similar: assets associated with socially disapproved activities may be exposed to higher financial and reputational risks, which can ultimately affect their pricing and risk profile.

2.7 Research Gap and Positioning

The literature reviewed above points to several gaps that this study aims to address. First, most research on ESG and credit risk focuses on non-financial firms rather than banks (Hauch, 2026; Apergis et al., 2022). Empirical evidence at the bank level remains comparatively limited, despite the important role banks play in financing economic activity and transmitting sustainability-related risks through their lending portfolios.

Second, much of the existing literature relies on market-based measures of credit risk, such as CDS spreads or bond yields, rather than realised credit outcomes like non-performing loan ratios. Market-based indicators capture forward-looking expectations, but they are also influenced by factors such as liquidity conditions, market sentiment, and investor behaviour. By contrast, NPL ratios provide a more direct measure of the quality of a bank’s loan portfolio, even though they are inherently backward-looking.

Third, the mechanism through which ESG affects bank credit risk remains insufficiently explored. Existing studies often focus either on firm-level ESG performance or on broad market reactions, but rarely distinguish between the role of a bank’s own ESG profile and the role of its sectoral lending exposure. Our two-model framework is designed to address this issue directly by separating general ESG performance from transition-risk exposure linked to lending portfolios.

Another important challenge concerns ESG measurement itself. The literature highlights substantial problems related to ESG rating divergence and measurement error (Berg et al., 2022, 2024). Different ESG providers frequently produce inconsistent ratings for the same firm, raising concerns about the reliability of ESG scores as indicators of underlying sustainability risk. These measurement issues may partly explain the mixed empirical evidence observed across previous studies.

In addition, prior research suggests that sustainability effects in credit markets depend strongly on the level of analysis. Hoepner et al. (2016) find that country-level sustainability has a stronger effect on borrowing costs than firm-level ESG measures. Similarly, institutional

evidence from the ECB (ECB, 2022) highlights sectoral exposure to carbon-intensive industries as a key transmission channel for climate-related financial risk.

Finally, Carnevale & Drago (2024) show that banks do incorporate ESG risks into lending decisions, but the existing evidence is based largely on loan pricing rather than realised credit outcomes. This points to the need for more granular approaches that focus directly on measures such as non-performing loans.

Our study addresses these gaps through two complementary empirical models. Model 1 follows the established ESG literature and examines the relationship between ESG scores and bank credit risk. Model 2 extends this approach by introducing the REEI index, which captures transition-risk exposure based on the sectoral composition of bank loan portfolios. Together, these two approaches provide a broader and more detailed perspective on how sustainability considerations may influence bank credit risk.

2.8 Hypotheses

Building on the theoretical arguments and empirical evidence discussed above, this study develops a set of testable hypotheses regarding the relationship between ESG performance, sectoral transition risk, and bank credit risk. While the existing literature provides mixed evidence on the role of ESG in explaining credit risk, it consistently highlights the importance of sustainability-related factors, measurement challenges, and the potential relevance of sectoral exposure. In particular, the combination of ESG-based measures and sector-level transition risk indicators allows for a more comprehensive assessment of how sustainability considerations affect bank loan portfolio quality.

ESG performance and credit risk. A large body of literature suggests that ESG performance is negatively associated with financial risk. From a theoretical perspective, strong ESG practices can improve risk management, reduce exposure to environmental and social controversies, and enhance transparency and stakeholder trust. These mechanisms reduce information asymmetry and expected default risk, thereby lowering credit risk. Empirical evidence supports this view. Liu & Xie (2024) show that higher ESG performance is associated with lower bank risk, partly through improvements in asset quality and reductions in non-performing loans (NPLs). Similarly, Al-Qudah et al. (2022) find that a higher share of green lending is associated with lower NPL ratios, indicating that sustainability-oriented lending improves loan portfolio quality. However, the literature also highlights important limitations. Hoepner et al. (2016) find that firm-level ESG performance does not have a robust effect on loan pricing, suggesting that aggregated ESG scores may not fully capture firm-specific risk. In addition, measurement error and rating divergence (Berg et al., 2022, 2024) may attenuate estimated ESG effects. Despite these limitations, the theoretical arguments and empirical evidence suggest that ESG performance should be negatively related to realised credit risk in the banking context.

Hypothesis 1 (H1). Higher ESG performance is associated with lower bank credit risk, as measured by non-performing-loan (NPL) ratios.

Sectoral transition risk and credit risk. While ESG scores provide a broad measure of sustainability performance, they may fail to capture specific risk channels that are particularly relevant for banks. Climate transition risk, arising from policy changes, technological disruption, and shifts in market preferences, is highly sector-specific and primarily transmitted through banks’ lending portfolios. Institutional evidence from the ECB (ECB, 2022) shows that banks are heavily exposed to carbon-intensive sectors, which are particularly vulnerable to transition risk. Similarly, Battiston et al. (2017) demonstrate that financial institutions’ exposures to climate-policy-relevant sectors represent a substantial share of their balance sheets, particularly through lending activities. From a theoretical perspective, banks with higher exposure to carbon-intensive sectors are more vulnerable to credit deterioration as these sectors face regulatory tightening, declining asset values, and increased default risk during the transition to a low-carbon economy. To capture this channel, this study introduces the Rated ESG Exposure Index (REEI), which weights each bank’s sectoral loan composition by climate-transition risk weights derived from Battiston et al. (2017).

Hypothesis 2 (H2). Higher exposure to carbon-intensive sectors, as measured by the REEI, is associated with higher bank credit risk (NPL ratios).

Relative importance of the two channels. The existing literature suggests that aggregated ESG scores may not fully capture the relevant dimensions of credit risk, particularly in the banking sector. Hoepner et al. (2016) find only limited evidence that firm-level ESG performance affects borrowing costs, while Berg et al. (2024) show that ESG ratings are subject to substantial measurement error and rating divergence across providers. These issues raise concerns about the reliability of ESG scores as indicators of underlying financial risk. At the same time, the institutional and climate finance literature emphasises the importance of sectoral exposure as a key transmission channel for risk. In particular, banks with large exposures to carbon-intensive industries may face higher transition-related credit risk as climate policies tighten and economic activity shifts toward lower-carbon sectors.

This raises the question of whether sector-based measures of transition risk provide stronger explanatory power than conventional ESG scores. Given that the REEI index directly captures banks’ exposure to high-risk sectors through the composition of their lending portfolios, it may provide a more precise and economically meaningful measure of sustainability-related credit risk than aggregate ESG indicators alone.

Hypothesis 3 (H3). Sector-based transition-risk measures (REEI) provide stronger explanatory power for bank credit risk than aggregate ESG scores.

3. Methodology and Data

3.1 Research Design Overview

Our empirical framework employs a two-model structure designed to test the ESG–credit risk association through two complementary channels. Model 1 examines the broad ESG signal: it

tests whether banks with higher ESG ratings from Bloomberg exhibit lower NPL ratios, controlling for bank-level financial characteristics and macroeconomic conditions. This is the main specification and covers the full sample period (2015–2024) with 35 banks, yielding a panel of 350 bank-year cells (314 to 347 observations after listwise deletion, depending on which ESG measure enters the regression). Model 2 examines the balance-sheet mechanism: it tests whether banks whose lending is concentrated in high-transition-risk sectors (as measured by the REEI) experience higher NPL ratios, using a smaller subsample of 32 banks for which Pillar 3 sectoral data are available over 2018–2024 (151 bank-year observations after listwise deletion). Both models use the same econometric framework (two-way fixed effects with robust standard errors), the same dependent variable (NPL ratio), and the same control variables, differing only in the key explanatory variable. Section 4.6 additionally reports a horse-race specification that estimates both models on the identical 149-observation overlap sample to test whether the perception and mechanism channels capture overlapping or independent variation.

The choice of a two-model structure rather than a single model that includes both ESG scores and REEI simultaneously is deliberate. The two variables measure conceptually distinct aspects of a bank’s ESG profile: the ESG score is an external perception metric assigned by a rating agency, while the REEI is a researcher-constructed measure of actual portfolio composition. Including both in a single regression would risk multicollinearity (if the two are correlated, which is plausible) and would obscure the separate channels. Moreover, the two models operate on different samples (full sample vs. Pillar 3 subsample) and different time windows, making a combined specification impractical.

3.2 Sample Construction

The sample comprises 35 listed European commercial banks drawn from the following 17 sample countries: France, Germany, Spain, Italy, the Netherlands, the United Kingdom, Switzerland, the Nordic countries (Sweden, Norway, Denmark, Finland), Austria, Belgium, Ireland, Portugal, Poland, and Greece. Banks were selected based on three criteria: (i) listing on a European stock exchange (to ensure availability of Bloomberg ESG coverage), (ii) classification as a commercial bank or universal bank with a material lending portfolio (excluding pure investment banks and custody banks), and (iii) availability of NPL ratio data for at least five consecutive years within the 2015–2024 window. The sample is intentionally broad in country coverage to capture heterogeneity in regulatory environments, banking market structures, and macroeconomic conditions, all of which are controlled for through bank fixed effects (which absorb country effects, since country does not vary within bank) and year fixed effects (which absorb common time shocks).

The Model 2 subsample is smaller, comprising the banks for which we were able to extract consistent Pillar 3 sectoral loan data at the NACE sector level. The Model 2 panel is constructed exclusively from CQ5-format disclosures, which became standardised under the EBA Implementing Technical Standards on Pillar 3 ESG disclosures in June 2022 (applicable from the 2022 reporting year for large institutions). We restrict the sample to CQ5-equivalent bank-years for two reasons rooted in internal validity. First, pre-CQ5 disclosures (typically the CR1-B template) use bank-specific or country-specific sectoral taxonomies that do not map cleanly to the standardised NACE Rev2 breakdown required for the REEI calculation; reconciling them would

introduce non-trivial measurement noise that varies systematically across banks. Second, CR1-B reports loans on a broader scope that bundles households, sovereigns, and financial counterparties into the same template, so the denominator from which we compute sectoral shares is not directly comparable across the two formats. Mixing the two regimes would weaken the homogeneity of the regressor and the comparability of REEI values across the panel. Restricting to CQ5 preserves a uniform sectoral taxonomy and a uniform exposure scope across all bank-years, at the cost of a smaller and unbalanced panel.

After hand-collecting and reconciling CQ5-format Pillar 3 disclosures across the full bank list, the Model 2 sample comprises 32 of the 35 banks observed over 2018–2024, with 151 bank-year observations after listwise deletion on the regression variables. Coverage is sparse in 2018–2019 (11 of the 151 observations) because most banks did not voluntarily adopt the CQ5 format before it became mandatory; coverage thickens substantially from 2020 onward, with 32 banks reporting in each year from 2022 to 2024. Three banks from the Model 1 panel are excluded from Model 2 for data-availability reasons. *UBS Group* is not subject to the EU CRR Pillar 3 ESG ITS in the same way as EU SSM banks; while UBS does publish a Pillar 3 report under FINMA rules, its sectoral exposure breakdown follows a different taxonomy that cannot be reconciled with CQ5 without strong assumptions. *Credit Suisse* ceased to exist as an independent entity following its emergency acquisition by UBS in March 2023, and its standalone disclosures do not provide CQ5-equivalent data over the relevant window. *Millennium BCP* does not publish CQ5 in the form used by the rest of the sample. We treat the exclusion of these three banks as preferable to admitting heterogeneous proxies on a per-bank basis, which would shift the empirical exercise from analysis to data engineering and undermine the comparability of the regressor across the panel. The Model 2 sample is summarised in Section 3 alongside the variable definitions; the full bank list (Table 1) and the Pillar 3 extraction notes (Appendix A) document scope and reconciliation issues bank by bank.

Table 1: Sample Banks

#	Ticker	Bank Name	Country	Notes
1	BNP FP Equity	BNP Paribas	France	
2	ACA FP Equity	Crédit Agricole	France	
3	GLE FP Equity	Société Générale	France	
4	DBK GR Equity	Deutsche Bank	Germany	
5	CBK GR Equity	Commerzbank	Germany	
6	SAN SM Equity	Banco Santander	Spain	
7	BBVA SM Equity	BBVA	Spain	
8	CABK SM Equity	CaixaBank	Spain	
9	UCG IM Equity	UniCredit	Italy	
10	ISP IM Equity	Intesa Sanpaolo	Italy	
11	BAMI IM Equity	Banco BPM	Italy	
12	INGA NA Equity	ING Group	Netherlands	
13	ABN NA Equity	ABN AMRO	Netherlands	IPO 20 Nov 2015; FY 2015 included
14	UBSG SW Equity	UBS Group	Switzerland	
15	NDA SS Equity	Nordea	Finland	
16	DANSKE DC Equity	Danske Bank	Denmark	AML scandal; dummy 2018–2019
17	SEBA SS Equity	SEB	Sweden	
18	SWEDA SS Equity	Swedbank	Sweden	AML scandal; dummy 2019
19	DNB NO Equity	DNB	Norway	
20	HSBA LN Equity	HSBC Holdings	UK	
21	LLOY LN Equity	Lloyds Banking Group	UK	
22	BARC LN Equity	Barclays	UK	
23	NWG LN Equity	NatWest Group	UK	
24	EBS AV Equity	Erste Group	Austria	
25	RBI AV Equity	Raiffeisen Bank Intl	Austria	Russia exposure post-2022 (descriptive flag, no dummy)
26	CSGN SW Equity	Credit Suisse	Switzerland	Distress; dummy 2020–2023; data truncated after 2023
27	KBC BB Equity	KBC Group	Belgium	
28	EUROB GA Equity	Eurobank	Greece	
29	PKO PW Equity	PKO Bank Polski	Poland	
30	PEO PW Equity	Bank Pekao	Poland	
31	SAB SM Equity	Banco Sabadell	Spain	
32	BMPS IM Equity	Monte dei Paschi di Siena	Italy	Distressed 2016–2019 (descriptive flag, no dummy)
33	AIBG ID Equity	AIB Group	Ireland	
34	BIRG ID Equity	Bank of Ireland	Ireland	
35	BCP PL Equity	Millennium BCP	Portugal	

Notes: The sample comprises 35 listed European commercial banks. Bloomberg ticker codes follow the standard [Ticker] [Country] Equity convention. The Notes column flags banks with idiosyncratic events or unusual treatment in the panel. The Credit Suisse, Danske Bank, and Swedbank events are absorbed by dedicated event dummies in the regression (see Section 3.6.4); other notes (Monte dei Paschi 2016–2019 distress, Raiffeisen Russia exposure, ABN AMRO IPO date) are descriptive flags whose effects are absorbed by the bank and year fixed effects without a separate event dummy. Banks without notes have no special treatment. All banks are observed over the full sample period (2015–2024) except where noted.

3.3 Dependent Variable: NPL Ratio

3.3.1 Choice of credit risk measure

The dependent variable in both models is the non-performing loan (NPL) ratio, defined as the stock of non-performing loans divided by gross loans. Several alternative credit risk measures were considered before arriving at this choice, and the reasoning behind the selection merits explicit discussion.

The most common alternatives in the ESG–credit risk literature are credit default swap (CDS) spreads, bond yield spreads, probability of default (PD) measures, and loan loss provisions. CDS spreads, used by Hauch (2026) in her study of 562 non-financial firms, capture market-implied credit risk and have the advantage of being forward-looking and available at high frequency. However, CDS spreads measure the market’s assessment of the bank’s own default risk, not the credit quality of the bank’s lending portfolio. A bank can have low CDS spreads (because the market considers it unlikely to default) while simultaneously holding a deteriorating loan book, if the bank is well-capitalised or has implicit government support. Since our research question asks about the quality of the lending portfolio itself, not the market’s view of the bank as a whole, CDS spreads are not the appropriate measure for our purpose.

Bond yield spreads, used by Apergis et al. (2022), suffer from similar limitations: they incorporate liquidity premia, maturity risk, and issuer-specific factors beyond credit quality. Probability of default measures, while conceptually attractive, are typically model-based outputs (e.g., from Merton-type structural models or from rating agency proprietary models) that introduce an additional layer of estimation uncertainty. Loan loss provisions under IFRS 9 are forward-looking and capture Expected Credit Losses (ECL), but provisioning practices vary significantly across banks depending on their internal models, risk appetite, and regulatory interactions, making cross-bank comparisons difficult.

The NPL ratio, while backward-looking and subject to accounting regime changes, offers three advantages for our study. First, it directly measures realised credit deterioration in the lending portfolio, which is the outcome variable most aligned with our research question. Second, it is widely reported, consistently defined across European banks since the EBA’s harmonisation of the NPL definition in 2014, and available from Bloomberg for all sample banks over the full sample period. Third, it is an outcome variable (realised credit losses) rather than a market perception variable (spreads or ratings), which is important for disentangling whether ESG performance actually predicts credit outcomes or merely influences market perceptions of risk.

3.3.2 Definition and data source

We define the NPL ratio as:

$$\text{NPL Ratio}_{i,t} = \frac{\text{Non-Performing Loans}_{i,t}}{\text{Gross Loans}_{i,t}}. \quad (1)$$

Non-performing loans follow the EBA’s harmonised definition: exposures where the borrower is unlikely to pay in full without realisation of collateral, or where the borrower is more than 90 days past due on any material credit obligation. Gross loans are total on-balance-sheet lending before

deduction of provisions. Data are sourced from Bloomberg (field: `NPL_RATIO`), cross-checked against banks’ annual reports and Pillar 3 disclosures where discrepancies arise. The variable is expressed as a percentage (e.g., 3.2% means non-performing loans equal 3.2% of gross loans).

We considered but decided against applying a logarithmic transformation to the NPL ratio. Log transformation is sometimes used to reduce right-skewness and stabilise variance, and the NPL ratio is indeed right-skewed in our sample (a few banks have very high NPL ratios, while most cluster below 5%). However, the NPL ratio has a natural lower bound of zero, and $\log(0)$ is undefined. Several bank-year observations have NPL ratios very close to zero (e.g., Nordic banks in expansion years), and adding an arbitrary constant before taking logs (e.g., $\log(\text{NPL} + 1)$) would distort the economic interpretation of the coefficients. We therefore estimate the model in levels, and use winsorisation at the 1st/99th percentiles as a robustness check to assess the sensitivity to extreme outliers.

3.3.3 The IFRS 9 structural break

A key methodological consideration is the IFRS 9 structural break. In January 2018, the International Financial Reporting Standard IFRS 9 replaced IAS 39 as the accounting standard for financial instruments in the European Union. The most significant change for NPL measurement is the shift from an “incurred loss” model (under IAS 39, provisions were recognised only when a loss event had already occurred) to an “expected loss” model (under IFRS 9, provisions are recognised based on forward-looking expectations of credit losses). This transition affected the timing and magnitude of loan loss provisions, which in turn can influence reported NPL ratios through the interaction between provisioning coverage and write-off policies. Banks that had been under-provisioning under IAS 39 may have experienced a one-time increase in provisions upon IFRS 9 adoption, potentially accompanied by changes in NPL classification thresholds.

We address the IFRS 9 break through two mechanisms. First, year fixed effects in the regression absorb any common level shift in NPL ratios across all banks in 2018. If the IFRS 9 transition caused a uniform increase (or decrease) in reported NPL ratios across the sample, the year dummy for 2018 will capture this effect, and the ESG coefficient will be estimated from the remaining within-year variation. Because IFRS 9 was adopted simultaneously by every bank in our sample on 1 January 2018, a separate “post-2018” indicator would be a perfect linear combination of the post-2018 year fixed effects, drop out of the regression on collinearity grounds, and contribute no identifying variation. We therefore deliberately do not include an IFRS 9 dummy as a control: the structural break is already fully absorbed by the year fixed effects. Second, we conduct a post-2018 subsample robustness check that restricts the sample to the 2018–2024 period, within which the accounting regime is homogeneous. If the ESG coefficient is qualitatively similar in the post-2018 subsample, this provides reassurance that the IFRS 9 transition does not drive the main results.

3.4 Model 1 Key Explanatory Variable: ESG Scores

3.4.1 Choice of ESG provider

The key explanatory variable in Model 1 is the Bloomberg ESG score. The choice of ESG data provider is consequential: as documented by Berg et al. (2022), ESG ratings from different providers correlate at only 0.54 on average, meaning that the same bank can receive very different ESG assessments depending on which provider is consulted. We selected Bloomberg over the main alternatives, Refinitiv, MSCI, Sustainalytics, and S&P Global, for three reasons.

First, Bloomberg provides the broadest and most consistent coverage of European listed banks over our sample period (2015–2024). Several alternative providers have significant coverage gaps for smaller European banks or for earlier years. Second, Bloomberg’s ESG methodology is based entirely on publicly disclosed information (annual reports, sustainability reports, regulatory filings), without relying on company questionnaires or subjective analyst assessments. This means the scores reflect disclosed rather than self-reported performance, reducing concerns about social desirability bias in reporting. Third, Bloomberg data were accessible to us through the ZHAW Bloomberg Terminal, ensuring that the ESG data could be extracted alongside the financial variables (NPL ratios, CET1, ROA, total assets) in a single, internally consistent data pull.

The decision to use a single provider follows the recommendation of Berg et al. (2022): when using ESG ratings to measure a specific company characteristic (in our case, as a proxy for the bank’s overall sustainability performance), the researcher should choose the provider whose methodology best fits the research question and clearly document this choice. The limitation, that our results are conditional on Bloomberg’s specific scoring methodology and may not generalise to other providers, is acknowledged explicitly in Section 5.4.

3.4.2 Score definition and scale

We use Bloomberg’s ESG *performance* scores: the combined ESG score (ESG_SCORE) and the three pillar scores (ENVIRONMENTAL_SCORE, SOCIAL_SCORE, GOVERNANCE_SCORE). All four scores are produced on a 0–10 scale, with higher scores indicating better management of financially material ESG issues relative to the company’s industry peers. We do not rescale or normalise the scores, using them as-is from Bloomberg. The variable is continuous and approximately normally distributed in our sample (see Section 4), which simplifies interpretation: a one-unit increase in the ESG score corresponds to a one-point improvement on the Bloomberg 0–10 scale.

It is important to distinguish the Bloomberg *performance* scores used here from the Bloomberg *disclosure* scores (ESG_DISCLOSURE_SCORE, ENVIRON_DISCLOSURE_SCORE, SOCIAL_DISCLOSURE_SCORE, GOVNCNCE_DISCLOSURE_SCORE), which are reported on a 0–100 scale and measure only the breadth of a company’s ESG reporting (how many ESG data points it publishes), not the quality of the underlying practices. The disclosure scores are widely available on the Bloomberg Terminal but are conceptually different from the performance scores and are not appropriate for our research question, which concerns whether *better-managed* banks have lower NPL ratios rather than whether *more-transparent* banks do. Throughout the remainder of the paper, “ESG score” refers to the performance score unless explicitly noted otherwise.

3.4.3 Pillar decomposition rationale

In addition to the aggregate ESG score, we estimate separate models using the Environmental (E), Social (S), and Governance (G) pillar scores individually. This decomposition is motivated by two considerations. First, the theoretical channels through which each pillar might affect credit risk differ. The Environmental pillar captures the bank’s management of climate and environmental risks, which, through the transition risk channel, could directly affect the creditworthiness of borrowers in carbon-intensive sectors. The Social pillar captures workforce practices and community relations, which may affect credit risk indirectly through operational stability and reputational capital. The Governance pillar captures board oversight, audit quality, and internal controls (factors that are directly relevant to loan origination standards and credit monitoring). If these channels operate with different intensities, the aggregate ESG coefficient blends them together, potentially masking a strong effect in one dimension with weak or zero effects in others.

Second, prior studies report different results across pillars. Hauch (2026) finds that only the Environmental pillar drives the ESG divergence–credit risk relationship; the Social and Governance pillars are not statistically significant. Apergis et al. (2022), in contrast, find significance across all three pillars, with the Social pillar showing the strongest coefficient. By testing each pillar separately in our bank-level setting, we can contribute to this ongoing debate and assess whether the pillar-level patterns observed for non-financial corporates extend to banks.

3.4.4 Bloomberg score architecture

The Bloomberg ESG performance score has a multi-layer construction that is worth describing in some detail, both because the methodology has implications for how the coefficient estimates in Section 4 should be interpreted and because the architecture is non-trivially different from competing rating systems (e.g., MSCI, Sustainalytics, Refinitiv).¹ The score is built from the bottom up in five stages.

Stage 1 – Field Scores. The lowest building block is a *field*: a single sustainability data item disclosed by the company (e.g., scope-1 emissions, percentage of women on the board, presence of a climate transition policy). Each field is scored on a 0–10 scale using a parametric scoring curve estimated from the empirical distribution of that field within the company’s peer group. The peer group is defined by Bloomberg’s proprietary ESG Classification System (BECS), which groups firms exposed to similar ESG issues based on business model and industry characteristics rather than on standard industry codes. Field scores reward both absolute performance and relative position within the peer group.

Stage 2 – Sub-Issue and Issue Scores. Field scores are aggregated up into Sub-Issue Scores (e.g., “GHG emissions intensity”), and Sub-Issue Scores into Issue Scores (e.g., “Climate Change”). Aggregation uses a weighted shifted p -mean with $p = 0.5$, which rewards consistent performance across multiple sub-issues and penalises uneven performance.

Stage 3 – Disclosure Factor adjustment. A bank-specific Disclosure Factor (DF) between 0 and 1 is computed at the Issue level, capturing the fraction of relevant fields for which the bank

¹The description below summarises Bloomberg’s published methodology document (Bloomberg, 2025). The white paper is updated periodically and the version cited here is the most recent at the time of writing.

actually discloses quantitative or non-policy data. The DF is then applied to the Performance Score through an Upper Target function $\text{Upper Target} = 10 \times (0.9 \times \text{DF}^2 + 0.1)$, which scales the Issue Score into a disclosure-driven range. A bank with zero quantitative disclosure cannot achieve an Issue Score above 1.0; a bank with full disclosure can achieve the full 0–10 range. This design embeds a disclosure incentive directly into the performance score: improvements in reporting mechanically raise the upper bound of the score, even before any change in underlying practices.

Stage 4 – Pillar Scores. Issue Scores are aggregated into the three Pillar Scores (E, S, G) using industry-specific materiality weights. For banks, the materiality weighting emphasises issues such as climate strategy, governance of financed emissions, lending portfolio ESG risk, business ethics, and data privacy. The pillar weights are not equal-weighted across issues but reflect Bloomberg’s assessment, informed by the SASB framework and Bloomberg Intelligence research, of which issues drive financial materiality for the banking sector specifically.

Stage 5 – Combined ESG Score. The three Pillar Scores are combined into a single ESG Score using sector-specific pillar weights. For banks, the Governance pillar typically receives the largest weight, reflecting the systemic importance of bank governance and the relative stability of governance metrics over time.

Two features of this architecture are particularly relevant to our regression interpretation. First, because the Disclosure Factor enters multiplicatively at the Issue level, the performance score and the disclosure score are mechanically correlated even though they are conceptually distinct fields. A bank that improves its disclosure breadth will, holding underlying practices fixed, see its performance score rise. This is by design: Bloomberg (2025) note that the construction is intended to incentivise disclosure of quantitative ESG data, even when that disclosure reveals poor performance. Second, the BECS (Bloomberg ESG Classification System) peer-group benchmarking means that scores are not directly comparable across peer groups. A score of 5 in a peer group with high average disclosure is not equivalent to a score of 5 in a peer group with lower average disclosure. Because European listed commercial banks form a relatively homogeneous BECS peer group, this concern is mitigated within our sample, but it remains a caveat for any cross-industry comparison.

3.5 Model 2 Key Explanatory Variable: The REEI

3.5.1 Conceptual motivation

The Rated ESG Exposure Index (REEI) is a composite measure constructed to capture the climate transition risk embedded in a bank’s lending portfolio. The motivation for constructing this variable is that aggregate ESG scores (as used in Model 1) measure the bank’s perceived ESG performance but do not directly measure the risk characteristics of its loan book. A bank can achieve a high ESG score through strong governance practices and workplace policies while simultaneously maintaining significant lending exposure to fossil fuel companies or other carbon-intensive sectors. The REEI is designed to capture this portfolio composition dimension that ESG scores may miss.

3.5.2 Construction formula

For each bank i in year t , the REEI is calculated as:

$$\text{REEI}_{i,t} = \sum_s \text{SectorShare}_{i,s,t} \times \text{CRW}_s, \quad (2)$$

where $\text{SectorShare}_{i,s,t}$ is the share of sector s in bank i 's gross loan portfolio in year t , and CRW_s is the Climate Risk Weight assigned to sector s . The summation runs over the five Battiston et al. (2017) climate-policy-relevant categories (fossil fuel, utilities, energy-intensive industries, transport, and housing) plus a residual “other” bucket that we add to accommodate sectors with limited direct climate-policy exposure. The six buckets together cover the 19 NACE Rev2 sector divisions. A bank with a higher REEI has a lending portfolio tilted toward sectors facing greater climate transition risk; a bank with a lower REEI lends predominantly to sectors with limited exposure to climate policy changes. The REEI is bounded between 0.05 (if 100% of loans were in the lowest-risk “other” bucket, which carries $\text{CRW} = 0.05$) and 1.0 (if 100% were in fossil fuels and mining, which carries $\text{CRW} = 1.0$); in practice observed values cluster in a much narrower range (sample mean 0.32, standard deviation 0.04, and a tight 0.23–0.41 spread across the 151 valid bank-years).

3.5.3 Pillar 3 data extraction process

The sectoral loan exposure data underpinning the REEI are hand-collected from banks' Pillar 3 regulatory disclosures. Under the Capital Requirements Regulation (CRR), large European banks are required to disclose a range of credit risk information in their Pillar 3 reports, including exposures by asset class and, increasingly, by sector. However, the format, granularity, and definitions used in these disclosures vary significantly across banks and across reporting years. We document the key extraction challenges below.

Format heterogeneity: Some banks (e.g., BNP Paribas, BBVA from 2020 onwards) report sectoral exposures using the 19-division NACE Rev2 classification (sectors A through S), which maps directly to our CRW framework. Others (e.g., Deutsche Bank and Commerzbank before 2021) report using broader or bank-specific sector taxonomies that do not correspond to NACE divisions, requiring manual mapping. For pre-2018 years, several banks report only the CR1-B template (breakdown by exposure class and asset type) without sector-level detail, which prevents REEI calculation for those years. We therefore restrict the Model 2 sample to years where NACE-level data are available, with coverage starting in 2018 for early-adopter banks (Italian and French banks lead) and broadening to most of the sample by 2021–2022 as banks transition to the EBA-format Pillar 3 templates. The Model 2 panel ultimately spans 2018–2024 with 151 bank-year observations across 32 banks, with most of the variation concentrated in the three dense-coverage years 2022–2024. Figure 1 visualises the bank-by-year coverage pattern; bank-specific extraction notes are provided in Appendix A.

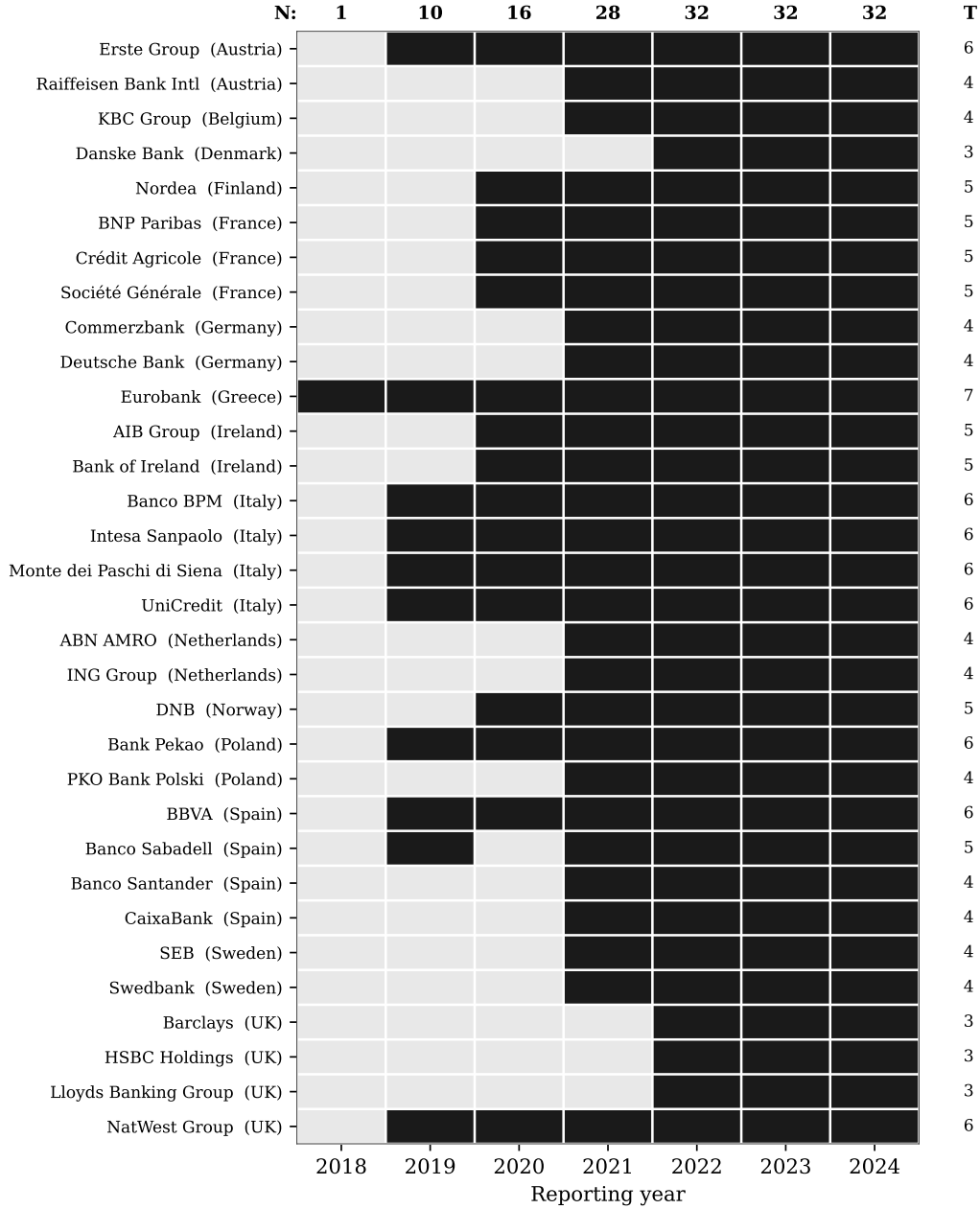


Figure 1: Pillar 3 CQ5 sectoral disclosure coverage in the Model 2 panel. Each filled cell indicates a bank-year for which a CQ5-equivalent NACE sectoral breakdown was available and the REEI could be computed. Banks are grouped by country. Column totals (top) report the number of banks reporting in each year; row totals (right) report the number of years each bank contributes. The visible diagonal pattern reflects the staggered adoption of the EBA-format Pillar 3 templates: the 2018–2019 cells are sparse (1 and 10 banks, respectively), 2020–2021 reflect the broadening of voluntary reporting, and the 2022–2024 cells are full (32 banks each year) once the EBA’s Implementing Technical Standards on Pillar 3 ESG disclosures became binding for large institutions. Total: 151 bank-years across 32 banks. Source: authors’ compilation from banks’ Pillar 3 reports.

Definition inconsistencies: CaixaBank reports “defaulted” exposure at the sector level rather than “non-performing” exposure. Under CRR definitions, “default” and “non-performing” are closely related but not identical concepts (the NPE definition is slightly broader). We treat

CaixaBank’s “defaulted” figures as a close proxy for NPL exposure but flag this discrepancy in Appendix A. For the REEI itself, the relevant input is the gross carrying amount by sector (not the NPL amount), so this issue affects only the calculation of sector-level NPL rates (which we report descriptively but do not use in the regression).

Sum-vs-total reconciliation: For each bank-year, we compare the sum of sector-level gross carrying amounts to the bank’s disclosed total gross loans. Discrepancies typically arise from differences in scope (e.g., the Pillar 3 sector breakdown may cover only corporate and SME lending, excluding retail mortgages) or from rounding. We record the reconciliation ratio for each bank-year and flag observations where the discrepancy exceeds 10%. These observations are retained in the sample but noted in the robustness discussion.

3.5.4 Climate Risk Weight (CRW) calibration

The Climate Risk Weights are calibrated on a 0.05–1.0 scale across six sector buckets (the five Battiston et al. (2017) climate-policy-relevant buckets plus a residual “other” bucket), into which the 19 NACE Rev2 sector divisions are mapped. The mapping reflects each sector’s vulnerability to transition risk based on three criteria, consistent with Battiston et al.’s approach: (i) direct GHG emissions intensity (Eurostat scope 1 CO₂ equivalent data), (ii) position in the fossil fuel supply chain (suppliers vs. users of fossil energy), and (iii) exposure to climate-related regulation (carbon pricing, emissions trading, energy efficiency mandates).

The highest weight (CRW = 1.0) is assigned to the fossil-fuel bucket, which contains NACE sector B (Mining and Quarrying) and includes fossil-fuel extraction, the activity most directly exposed to stranded-asset risk under decarbonisation scenarios. Energy-intensive industry (NACE C, Manufacturing) receives CRW = 0.7, reflecting the heavy reliance of materials production (steel, cement, chemicals) on fossil-fuel inputs. Transport (NACE H) receives CRW = 0.6, capturing the substantial transition risk in road haulage, aviation, and shipping. Utilities (NACE D, Electricity, Gas, Steam and Air Conditioning Supply) receive CRW = 0.5, reflecting the sector’s dual exposure: fossil-fuel-based utilities face significant stranded-asset risk, while renewable-based utilities may benefit; since the NACE 1-letter classification does not distinguish between them, we assign a moderate weight reflecting the blended risk. Housing (NACE F Construction and L Real Estate) receives CRW = 0.35, capturing energy-efficiency upgrade requirements and stranded-asset risk on energy-inefficient buildings under tightening EU performance standards. The residual “other” bucket, which includes services, public administration, finance, health, and education (NACE A, E, G, I, J, K, M, N, O, P, Q, R, S), receives CRW = 0.05, reflecting these sectors’ minimal direct climate-policy exposure. The full mapping is reported in Table 2.

Table 2: NACE-to-CRW Mapping

NACE	Sector name	Battiston bucket	CRW
A	Agriculture, forestry and fishing	Other	0.05
B	Mining and quarrying	Fossil fuel	1.00
C	Manufacturing	Energy-intensive	0.70
D	Electricity, gas, steam and air conditioning supply	Utilities	0.50
E	Water supply	Other	0.05
F	Construction	Housing	0.35
G	Wholesale and retail trade	Other	0.05
H	Transport and storage	Transport	0.60
I	Accommodation and food service activities	Other	0.05
J	Information and communication	Other	0.05
K	Financial and insurance activities	Other	0.05
L	Real estate activities	Housing	0.35
M	Professional, scientific and technical activities	Other	0.05
N	Administrative and support service activities	Other	0.05
O	Public administration and defence	Other	0.05
P	Education	Other	0.05
Q	Human health services and social work	Other	0.05
R	Arts, entertainment and recreation	Other	0.05
S	Other services	Other	0.05

Notes: Mapping from the 19 NACE Rev2 sector divisions to six sector buckets (the five Battiston et al. (2017) climate-policy-relevant buckets plus a residual “other” bucket), with the Climate Risk Weight (CRW) assigned to each bucket. Weights are author-calibrated based on the criteria described in the main text and informed by the climate-policy-relevance reclassification of Battiston et al. (2017). The mapping is applied unchanged across all bank-years in the Model 2 sample. Sensitivity to the weighting scheme is tested in the robustness checks of Section 4.5.

3.5.5 Econometric considerations of the REEI as a constructed variable

The REEI is a researcher-constructed variable, which raises several econometric considerations that we address explicitly. First, measurement error: the CRW weights involve subjective calibration choices, and the Pillar 3 data contain known inconsistencies (as documented above). In classical measurement error theory, measurement error in the explanatory variable biases the coefficient toward zero (attenuation bias), meaning our Model 2 estimates may understate the true relationship between portfolio transition risk and NPL ratios. Second, the REEI captures only the sectoral composition of the loan book and does not reflect within-sector credit quality: a bank lending to high-quality borrowers in a high-transition-risk sector will have the same REEI as a bank lending to weak borrowers in the same sector. This is a meaningful limitation. Third, the CRW weights are static (they do not vary over time), whereas the actual transition risk of a sector may evolve as climate policies tighten or technological disruptions shift the risk landscape. We address sensitivity to weight choices by testing alternative CRW specifications as a robustness check.

3.6 Control Variables

The selection of control variables follows a structured process designed to balance two competing concerns: including enough controls to mitigate omitted variable bias (which would inflate the ESG coefficient if omitted variables are positively correlated with ESG and negatively correlated with NPL) while avoiding the inclusion of redundant or collinear controls that would

inflate standard errors and reduce the precision of the key estimate. We group the controls into three categories: bank-level financial characteristics, country-level macroeconomic conditions, and event dummies for idiosyncratic episodes. Each variable is discussed below with its rationale, expected sign, data source, and any transformations applied.

3.6.1 Bank-level financial controls

All bank-level financial controls described below are sourced from Bloomberg, consistent with the data-provider choice for the ESG variables in Section 3.4.

Common Equity Tier 1 (CET1) ratio. The CET1 ratio is defined as the bank’s Common Equity Tier 1 capital divided by its risk-weighted assets, expressed as a percentage. It is the most important regulatory capital ratio in the Basel III framework and serves as a measure of the bank’s loss-absorbing capacity. We include CET1 as a control because capital adequacy and credit quality are jointly determined by the bank’s risk management decisions. Banks with higher CET1 ratios have a larger buffer to absorb loan losses, which may reduce the pressure to classify loans as non-performing (since the bank can afford to be patient with troubled borrowers). Conversely, weakly capitalised banks may engage in riskier lending to generate returns, leading to higher NPL ratios over time. The expected sign on CET1 is negative: better-capitalised banks are expected to have lower NPL ratios. No transformation is applied; the variable is used in its natural percentage form.

Return on Assets (ROA). ROA is defined as net income divided by total assets, expressed as a percentage. It measures the bank’s overall profitability relative to its asset base. We include ROA as a control for two reasons. First, profitability and credit quality are mechanically linked: loan write-offs reduce net income (and hence ROA), while profitable banks can afford more conservative credit policies. Second, ROA may be correlated with ESG performance (profitable banks have more resources to invest in ESG initiatives), and omitting ROA could create upward bias in the ESG coefficient, making ESG appear more beneficial than it actually is. The expected sign on ROA is negative: more profitable banks are expected to have lower NPL ratios, both because profitability allows for better risk management and because high NPL ratios directly reduce profitability. No transformation is applied.

Why ROA and not ROE? We initially considered both return on assets (ROA) and return on equity (ROE) as profitability measures. ROE is defined as net income divided by shareholders’ equity, and it is arguably a more intuitive measure of shareholder profitability. However, ROE is mechanically linked to leverage: a bank can have a high ROE simply by having high leverage (low equity relative to assets), independent of its operating performance. Since we already control for capital adequacy through CET1 (which is inversely related to leverage), including ROE alongside CET1 creates a near-mechanical relationship that inflates the variance inflation factor (VIF) above 5. Preliminary specifications including both ROA and ROE showed VIF values of 5.8 for ROE and unstable coefficient signs. We therefore retain ROA, which captures profitability independent of capital structure, and drop ROE.

Natural logarithm of total assets (LogTA). Bank size is measured as the natural logarithm of total assets in euros. We apply the logarithmic transformation for two reasons. First, the distribution of total assets across European banks is heavily right-skewed: the largest banks

in our sample (e.g., BNP Paribas, HSBC) have total assets exceeding €2 trillion, while smaller banks (e.g., Bank of Ireland, Millennium BCP) have total assets in the range of €50–200 billion. Using raw total assets would allow these extreme values to exert disproportionate influence on the regression estimates. The log transformation compresses the upper tail of the distribution and produces a more symmetric variable. Second, the log transformation implies a proportional rather than linear relationship between size and NPL: moving from €50 billion to €100 billion (a doubling) has the same estimated effect as moving from €500 billion to €1 trillion (also a doubling). This is more economically plausible than assuming that each additional euro of assets has the same marginal effect on NPL ratios regardless of the bank’s baseline size.

The expected sign on LogTA is ambiguous and depends on which size-related mechanism dominates. On the one hand, larger banks tend to have more diversified loan portfolios (across sectors, geographies, and borrower types), which should reduce concentration risk and lower NPL ratios. Larger banks also tend to have more sophisticated credit risk management systems. On the other hand, larger banks may be “too big to manage”: their complexity can lead to monitoring failures, especially in far-flung subsidiaries or specialised lending lines. The empirical banking literature finds mixed results for the size–NPL relationship.

A practical note on LogTA in a fixed effects setting: LogTA has very low within-bank variation, with only about 1.2% of the total variation in LogTA falling within-bank. Banks’ total assets change slowly over time (organic balance sheet growth of a few percent per year, with occasional jumps from acquisitions). This means the fixed effects model has limited ability to identify the coefficient on LogTA, because after subtracting each bank’s time-average LogTA, the residual variation is tiny. We retain LogTA in the model to absorb any marginal size-related variation in NPL ratios, but we do not interpret its coefficient substantively. Total assets are pulled directly in EUR from Bloomberg, which performs the currency conversion at end-of-period rates for non-EUR reporters (GBP, CHF, SEK, NOK, DKK, PLN); no manual conversion is applied.

3.6.2 Variables considered and dropped

In addition to the three bank-level controls retained in the final specification, we considered several other variables during the model development process. The following variables were tested and excluded, with the rationale documented for transparency.

Return on Equity (ROE) was dropped due to multicollinearity with CET1 and ROA, as discussed above ($VIF > 5$). Net Interest Margin (NIM), defined as net interest income divided by average interest-earning assets, was considered as a control for the bank’s business model intensity. Banks with higher NIM tend to be more focused on traditional lending (as opposed to fee-based or trading activities). However, NIM was moderately correlated with ROA (Pearson correlation of approximately 0.41) and added little explanatory power in preliminary regressions (its coefficient was small and insignificant in all specifications). Including it alongside ROA risked absorbing genuine variation in the ESG coefficient without adding meaningful control power. We therefore dropped NIM from the final specification.

Cost-to-income ratio (operating expenses divided by operating income) was considered as a measure of operational efficiency. Efficient banks may originate higher-quality loans (because they invest in better credit processes), but the cost-to-income ratio can also move mechanically

with revenue shocks (a decline in income raises the ratio even if costs are unchanged). In preliminary tests, the cost-to-income ratio was not significant and did not change the ESG coefficient materially. It was excluded to preserve degrees of freedom. Loan-to-deposit ratio (net loans divided by customer deposits) was considered as a measure of funding structure and liquidity risk. Banks that lend aggressively relative to their deposit base may face funding pressures that affect credit quality. However, this variable showed high within-bank correlation with LogTA (as balance sheets grow, both loans and deposits tend to grow proportionally), contributing to collinearity. It was excluded from the final specification.

Carbon emissions data (Scope 1+2 and carbon intensity) were initially extracted from Bloomberg as potential explanatory variables. However, these measures capture the bank’s own operational emissions (e.g., from office buildings and business travel), which are a tiny fraction of its financed emissions and bear little relationship to the credit risk of its lending portfolio. Including bank-level carbon emissions in a model that already includes ESG scores risks double-counting the environmental dimension and confusing the interpretation. We therefore exclude carbon emissions from the regression and note that the REEI (Model 2) provides a more appropriate measure of the bank’s climate-relevant portfolio exposure.

3.6.3 Country-level macroeconomic controls

Real GDP growth rate (annual, %). GDP growth is the most fundamental macroeconomic driver of credit quality. During economic expansions, borrowers generate higher revenues and incomes, enabling them to service their debts; during recessions, revenue declines and unemployment increases lead to higher default rates and rising NPL ratios. The expected sign is negative: higher GDP growth is associated with lower NPL ratios. This relationship is well-established in the macro-financial linkages literature and is a standard control in any study of bank credit risk. Data source: World Bank World Development Indicators (WDI), indicator NY.GDP.MKTP.KD.ZG, annual frequency. The local-currency basis ensures comparability across non-eurozone countries (UK, Switzerland, Sweden, Norway, Denmark, Poland) without EUR-denomination distortions.

Year-on-year change in 10-year sovereign bond yield (annual, percentage points). The change in long-term sovereign yields captures country-specific shifts in the funding and credit environment that bank borrowers face. Rising yields typically reflect monetary tightening, fiscal stress, or risk-premium shocks; falling yields the reverse. Borrower default rates respond to changes in the rate environment rather than to absolute yield levels, since bank fixed effects already absorb each country’s baseline yield level. The expected sign is positive: rising sovereign yields are associated with higher NPL ratios, both directly (higher debt-service costs for variable-rate borrowers and corporate borrowers refinancing) and indirectly (recessionary effects of monetary tightening). Data source: OECD Main Economic Indicators series “Long-term government bond yields, 10-year, main (including benchmark)”, retrieved via FRED (indicator IRLTLT01[XX]A156N where [XX] is the country code). The change is computed in the workbook as the year-on-year difference of the level series. The 2014 yield level is included as a baseline observation to compute the 2015 change.

Earlier specifications considered the unemployment rate as a separate macroeconomic control.

Unemployment was dropped on two grounds. First, in our European sample, GDP growth and unemployment exhibit substantial within-country correlation through Okun’s law (typically -0.6 to -0.8 in OECD countries), creating a redundancy that adds little explanatory power once GDP is controlled for. Second, the UK’s Office for National Statistics (ONS) changed its unemployment series methodology mid-sample (2017 reweighting), creating a structural break that would require splicing across two source definitions for one of the seventeen countries. Single-source consistency was prioritised; the year fixed effects absorb the labour-market component of common time variation, and the GDP control captures cross-country cyclical heterogeneity.

Real residential property price growth (annual, %). Year-on-year change in the inflation-adjusted residential property price index captures the housing-market channel through which credit risk transmits. Falling real house prices erode collateral value on mortgage loans, increase loan-to-value ratios on outstanding mortgages, and signal weakening household balance sheets. The 2022–2023 European real-house-price correction (driven by the inflation shock and ECB rate-hiking cycle) provides substantial within-sample variation in this variable: real prices fell by more than 10 percentage points in Germany (-13.6% in 2023), Sweden (-12.6%), Finland (-11.7%), and Austria (-8.8%), while remaining stable or rising in Greece, Switzerland, and Norway. The expected sign is negative: declining real house prices are associated with rising NPL ratios. Data source: BIS Residential Property Prices database (WS_SPP), series Q.[XX].R.771 (where [XX] is the country code), accessed 26 April 2026. BIS publishes this series at quarterly frequency only; we compute annual values as the simple arithmetic mean of the four quarterly observations within each calendar year, which captures the typical level of housing-market pressure throughout the year rather than a year-end snapshot.

Earlier specifications considered HICP (Harmonised Index of Consumer Prices) inflation as a separate control. Inflation was dropped on two grounds. First, the channel through which inflation affects credit risk is partially captured by the sovereign yield change variable, since central banks raise policy rates in response to inflation shocks, which feeds into long-term yields. Second, Eurostat’s HICP series has source breaks for several countries (UK left the EU and ONS uses a different methodology; Switzerland is not in HICP and uses a national CPI definition), preventing single-source consistency across all 17 sample countries. The Credit-to-GDP ratio (BIS) was also considered as an additional financial-cycle control but was dropped because it is slow-moving (largely absorbed by bank fixed effects) and BIS publishes it with a significant lag.

We note that the bank fixed effects absorb all time-invariant country characteristics (the average macroeconomic environment, institutional quality, legal system, banking market structure), while the year fixed effects absorb common time shocks that affect all countries simultaneously (e.g., the ECB’s monetary policy stance). The country-level macroeconomic controls therefore capture only the time-varying, country-specific component of macroeconomic conditions, the part that differs across countries within a given year and varies over time within a given country, net of the common time trend.

3.6.4 Event dummies

We include four event dummies to absorb idiosyncratic shocks that would otherwise contaminate the ESG coefficient estimate. Each dummy is motivated by a specific data observation during

the sample screening process.

CS_Distress: Equals one for Credit Suisse in the years 2020–2023, and zero otherwise. Credit Suisse experienced a series of risk management failures during this period (the Archegos Capital implosion in March 2021, the Greensill Capital collapse, and mounting concerns about the bank’s governance and risk culture) that culminated in its emergency acquisition by UBS in March 2023; we extend the dummy through 2023 to reflect the residual distress and write-down period during the takeover. The 2024 row for Credit Suisse is left blank in the panel, as the bank ceased to exist as an independent entity. These events caused extraordinary credit losses and NPL increases that are unrelated to the bank’s ESG score trajectory. Without this dummy, Credit Suisse’s deteriorating NPL ratios during 2020–2023 (a period in which its ESG score was also declining) could artificially strengthen the estimated ESG–NPL association by introducing an outlier that moves on both dimensions for idiosyncratic reasons.

AML_Danske and AML_Swedbank: Equal one for Danske Bank and Swedbank, respectively, in the years affected by their anti-money laundering (AML) scandals (Danske: 2018–2019, covering the disclosure and investigation of the Estonia branch case; Swedbank: 2019, the year of the SVT investigation and the consequent dismissal of CEO Birgitte Bonnesen). Both banks faced allegations of facilitating large-scale money laundering through their Baltic subsidiaries, resulting in regulatory penalties, management changes, and reputational damage. These scandals affected the banks’ financial metrics (including provisioning and NPL-related figures) in ways orthogonal to ESG performance as measured by Bloomberg’s scoring methodology. The dummies prevent these idiosyncratic events from biasing the ESG coefficient.

Ireland_GDP_2015: Equals one for Irish banks (e.g., AIB, Bank of Ireland) in 2015. In that year, Ireland’s real GDP growth was reported at 24.6%, an extraordinary figure driven by multinational corporations relocating intellectual property assets to Ireland for tax purposes (“leprechaun economics”). This statistical artefact distorts the GDP growth control variable for Irish banks in 2015, as the reported GDP growth does not reflect genuine domestic economic conditions. The dummy isolates this outlier from the macro control estimates.

3.6.5 Summary of variable choices

The full final regression specification can be summarised as follows. The dependent variable is the NPL ratio (Section 3.3). The key explanatory variable in Model 1 is the lagged Bloomberg combined ESG performance score, with the three pillar scores (Environmental, Social, Governance) used in the pillar-decomposition specification (Section 3.4); in Model 2 the key explanatory variable is the REEI (Section 3.5). The bank-level financial controls are the CET1 ratio, ROA, and the natural logarithm of total assets, with NIM, ROE, the cost-to-income ratio, the loan-to-deposit ratio, and Bloomberg-reported carbon emissions tested and dropped (Sections 3.6.1 and 3.6.2). The country-level macroeconomic controls are real GDP growth, the year-on-year change in the 10-year sovereign bond yield, and real residential property price growth, with unemployment, HICP inflation, and the credit-to-GDP ratio tested and dropped for redundancy or single-source-consistency reasons (Section 3.6.3). The event dummies are CS_Distress (Credit Suisse, 2020–2023), AML_Danske (Danske Bank, 2018–2019), AML_Swedbank (Swedbank, 2019), and Ireland_GDP_2015 (Irish banks, 2015) (Section 3.6.4). The IFRS 9 transition

is not coded as an event dummy because the regime change applies to all banks simultaneously in 2018 and is therefore absorbed by the year fixed effects (see Section 3.3). All variables enter the regression on their natural scale except total assets, which is log-transformed.

3.7 Econometric Specification

The baseline econometric model for both Model 1 and Model 2 takes the following two-way fixed effects form:

$$\text{NPL}_{i,t} = \alpha_i + \delta_t + \beta \cdot \text{ESG}_{i,t} + \gamma \cdot \mathbf{X}_{i,t} + \varepsilon_{i,t}, \quad (3)$$

where $\text{NPL}_{i,t}$ is the non-performing loan ratio of bank i in year t ; α_i is a bank-specific fixed effect that absorbs all time-invariant bank characteristics (including country, business model, and ownership structure); δ_t is a year fixed effect that absorbs all common time shocks (including the IFRS 9 transition, the COVID-19 pandemic, and interest rate regime changes); $\text{ESG}_{i,t}$ is the key explanatory variable (Bloomberg ESG score in Model 1, REEI in Model 2); $\mathbf{X}_{i,t}$ is a vector of control variables (CET1, ROA, LogTA, real GDP growth, change in 10-year sovereign bond yield, real residential property price growth, and event dummies); and $\varepsilon_{i,t}$ is the error term.

The coefficient β captures the within-bank, within-year association between ESG and NPL: it measures how much a bank’s NPL ratio changes, relative to its own time average and relative to the year average, when its ESG score changes by one unit. This within-variation is the identifying variation in a two-way fixed effects model. The bank fixed effect ensures that the coefficient is not driven by cross-sectional differences between high-ESG and low-ESG banks (which could reflect selection effects), and the year fixed effect ensures that the coefficient is not driven by common time trends (which could reflect market-wide shifts in both ESG scores and NPL ratios).

Standard errors are estimated using Huber–White heteroskedasticity-robust standard errors. We considered clustering standard errors at the bank level, which is the standard approach in panel data studies, but opted against it as the default specification because our cross-section of 35 banks is below the conventional threshold of 40–50 clusters needed for reliable cluster-robust inference (Cameron & Miller, 2015). With fewer clusters, the cluster-robust standard error estimator tends to under-reject the null hypothesis, producing standard errors that are too small. We include entity-clustered standard errors as a robustness check (Section 4.3) and report both sets of standard errors for transparency.

3.8 Robustness Framework

We conduct seven robustness checks designed to test the sensitivity of the main results to alternative specifications and sample definitions.

First, **contemporaneous ESG scores**: we replace the lagged baseline regressor $\text{ESG}_{i,t-1}$ with the same-year score $\text{ESG}_{i,t}$. Our baseline uses the lagged ESG score to ensure that the explanatory variable is predetermined relative to the dependent variable and to mitigate reverse causality (banks experiencing credit deterioration may simultaneously reduce ESG-related investments, which would inflate a contemporaneous negative estimate). The contemporaneous check tests whether the lag restriction is overly conservative, that is, whether the ESG–NPL association operates within a year and is therefore obscured by lagging. The check does not, however, resolve

endogeneity from time-varying omitted variables that simultaneously drive both variables.

Second, **post-2018 subsample**: we restrict the estimation to the 2018–2024 period, within which the IFRS 9 accounting regime is uniform across all sample banks. This check tests whether the main results are driven by the pre-2018 period, when NPL classification may have differed systematically under IAS 39.

Third, **excluding distressed banks**: we drop Credit Suisse, Danske Bank, and Swedbank from the sample. These banks experienced idiosyncratic events (CS collapse, AML scandals) that may distort the ESG–NPL relationship through outlier effects.

Fourth, **alternative standard error clustering**: we re-estimate the model with standard errors clustered at the bank (entity) level rather than using Huber–White robust standard errors, to test whether within-bank serial correlation affects inference.

Fifth, **individual ESG pillar decomposition**: we estimate Model 1 separately with Environmental, Social, and Governance pillar scores instead of the combined ESG score, to identify which dimension drives the overall association.

Sixth, **winsorised dependent variable**: we winsorise the NPL ratio at the 1st and 99th percentiles to mitigate the influence of extreme outliers. Some banks in the sample (particularly during the European sovereign debt crisis aftermath) had NPL ratios exceeding 15–20%, which could disproportionately influence the regression estimates.

Seventh, **lead-ESG placebo**: we regress $NPL_{i,t}$ on $ESG_{i,t+1}$, that is, on the one-period *lead* of the ESG score. Under a forward-looking interpretation of the lagged-ESG baseline, the lead coefficient should be small and statistically insignificant: ESG performance one year in the future should not predict NPL ratios in the current period. A large negative lead coefficient would indicate that rating-agency methodologies incorporate financial-stress signals into subsequent assessments, which would constrain the causal interpretation of the lagged-ESG baseline. We complement this combined-score test with a parallel **lead-S placebo** (R7s), which substitutes the Social-pillar score $S_{i,t+1}$ for $ESG_{i,t+1}$. Because the pillar decomposition (Section 4.2) shows that the Social pillar carries most of the credit-risk-relevant signal in our data, R7s diagnoses whether the same backward-looking re-rating dynamic that R7 detects on the combined score also operates within S, in which case the headline Social-pillar finding would need to be discounted accordingly.

3.9 Descriptive Statistics

Table 3: Descriptive Statistics (348 bank-year observations)

	N	Mean	SD	Min	P25	Median	P75	Max
NPL Ratio (%)	348	4.488	5.664	0.239	1.800	2.948	4.808	38.868
ESG Score (combined)	347	3.534	1.282	1.000	2.520	3.400	4.475	7.830
ESG Score (lagged)	314	3.359	1.192	1.000	2.473	3.200	4.167	6.910
Environmental pillar	312	2.252	1.956	0.040	0.470	1.745	3.900	7.260
Social pillar	347	3.207	1.676	0.380	1.950	2.910	4.105	8.730
Governance pillar	347	6.466	0.992	3.790	5.810	6.450	7.220	9.000
CET1 ratio (%)	348	14.802	2.481	8.000	12.995	14.300	16.400	25.000
ROA (%)	348	0.490	0.555	-2.396	0.280	0.485	0.780	2.269
Log(Total Assets)	348	12.983	1.064	10.585	12.208	13.081	13.788	14.885
GDP growth (%)	348	2.061	4.117	-10.940	0.976	2.067	3.306	24.624
Sov. yield 10y change (pp)	348	0.086	0.888	-2.382	-0.525	-0.121	0.343	4.109
Real RPP growth (%)	348	2.045	4.420	-13.567	-0.459	2.176	4.834	13.147

Descriptive statistics from the full sample (2015–2024) are reported in Table 3. The mean NPL ratio is 4.49%, with a standard deviation of 5.66% and substantial right-skew (median 2.95%, maximum 38.87% for distressed banks during the European sovereign debt crisis aftermath). The combined Bloomberg ESG score has a mean of 3.53 and a standard deviation of 1.28 on the 0–10 scale; the lagged version, which is our headline regressor, has a similar distribution with slightly lower mean (3.36, SD 1.19). The macroeconomic controls show the expected business-cycle range: GDP growth varies between -10.9% (pandemic-trough observations) and $+24.6\%$ (the Irish 2015 “leprechaun economics” artefact, isolated by the `Ireland_GDP_2015` dummy), the change in the 10-year sovereign yield spans -2.4 to $+4.1$ percentage points, and real residential property price growth ranges from -13.6% to $+13.1\%$.

Table 4: Within vs. Between Variation Decomposition

	Total SD	Within SD	Between SD	Within share
NPL Ratio (%)	5.664	4.056	4.010	0.716
ESG Score (combined)	1.282	1.079	0.699	0.841
ESG Score (lagged)	1.192	0.978	0.689	0.821
Environmental pillar	1.956	1.698	1.004	0.868
Social pillar	1.676	1.312	1.053	0.783
Governance pillar	0.992	0.558	0.832	0.562
CET1 ratio (%)	2.481	1.446	2.038	0.583
ROA (%)	0.555	0.453	0.326	0.817
Log(Total Assets)	1.064	0.133	1.068	0.125
GDP growth (%)	4.117	3.798	1.607	0.922
Sov. yield 10y change (pp)	0.888	0.883	0.099	0.994
Real RPP growth (%)	4.420	3.958	1.995	0.895

Notes: Total SD is the cross-sectional pooled standard deviation. Within SD is the standard deviation of $x_{i,t} - \bar{x}_i$ (deviation from each bank’s time average). Between SD is the standard deviation of bank-level time averages $\bar{x}_i$. Within share = Within SD / Total SD; small values indicate that bank fixed effects absorb most of the variation, leaving little within-bank variation for the FE estimator to identify the coefficient.

Table 4 decomposes the standard deviation of each variable into its within-bank and between-

bank components. This decomposition is informative for interpreting the fixed-effects regressions that follow: bank fixed effects absorb all between-bank variation, so only the within-bank variation, captured by Within SD, is available to identify the regression coefficients. The Within share column reports the fraction of total variation that is within-bank.

Two patterns are worth highlighting. First, the time-varying macroeconomic controls have very high within-bank shares (0.89–0.99), as expected: GDP growth, sovereign yield changes, and house-price growth are by construction time-series variables that move with the calendar rather than across cross-sectional units within a year. Second, LogTA exhibits the lowest within-bank share in the panel (0.125): bank size changes slowly over time, so almost 88% of its variation is between banks and is absorbed by the bank fixed effects. This explains why LogTA’s regression coefficient is imprecisely estimated and we do not interpret it substantively (Section 3.6.1). The combined ESG score (within share 0.82) and the Social pillar (0.78) sit comfortably in the middle of the distribution, while the Governance pillar (0.56) has noticeably less within-bank variation, consistent with governance practices being relatively stable within a bank over time.

3.10 Correlation Matrix

Table 5 reports pairwise Pearson correlations among the dependent variable, the ESG and pillar scores, the bank-level financial controls, and the macroeconomic controls. The variable numbering used in the matrix is given in the legend below the table.

Three patterns are worth noting. First, the NPL ratio is negatively correlated with the lagged ESG score ($r = -0.35$) and with each of the three pillar scores, with the largest pillar correlation on the Environmental pillar ($r = -0.33$) and the Governance pillar ($r = -0.32$). These cross-sectional correlations are directionally consistent with Hypothesis 1, although they reflect both within-bank and between-bank variation; the regression analysis (Section 4) uses the within-bank variation only. Second, the lagged ESG score is strongly correlated with the Environmental ($r = 0.77$) and Social ($r = 0.71$) pillar scores, and more weakly with the Governance pillar ($r = 0.43$). This is consistent with the Bloomberg combined ESG score being a sector-weighted aggregation of the three pillars, with the Governance pillar typically receiving a smaller relative weight for banks because Governance scores have less within-bank time variation. Third, none of the bank-level controls (CET1, ROA, LogTA) or macro controls correlate at $|r| > 0.5$ with any of the ESG variables, indicating no obvious multicollinearity threats. The largest control-to-control correlation is between LogTA and the Governance pillar ($r = 0.47$), reflecting the well-documented size-disclosure relationship in the ESG literature (Kotsantonis & Serafeim, 2019).

Table 5: Pairwise Correlation Matrix

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
(1)	1.000	-0.349	-0.329	-0.187	-0.317	-0.167	-0.304	-0.386	0.054	-0.191	-0.032
(2)	-0.349	1.000	0.766	0.710	0.426	0.042	0.195	0.341	-0.001	0.366	-0.315
(3)	-0.329	0.766	1.000	0.334	0.388	-0.019	0.142	0.427	0.003	0.356	-0.307
(4)	-0.187	0.710	0.334	1.000	0.130	0.037	0.245	-0.040	-0.011	0.263	-0.135
(5)	-0.317	0.426	0.388	0.130	1.000	0.039	-0.064	0.471	-0.003	0.139	-0.117
(6)	-0.167	0.042	-0.019	0.037	0.039	1.000	0.313	-0.356	0.077	0.061	0.041
(7)	-0.304	0.195	0.142	0.245	-0.064	0.313	1.000	-0.207	0.229	0.131	-0.021
(8)	-0.386	0.341	0.427	-0.040	0.471	-0.356	-0.207	1.000	-0.170	0.064	-0.192
(9)	0.054	-0.001	0.003	-0.011	-0.003	0.077	0.229	-0.170	1.000	0.149	0.240
(10)	-0.191	0.366	0.356	0.263	0.139	0.061	0.131	0.064	0.149	1.000	-0.423
(11)	-0.032	-0.315	-0.307	-0.135	-0.117	0.041	-0.021	-0.192	0.240	-0.423	1.000

Variables: (1) NPL Ratio (%); (2) ESG Score (lagged); (3) Environmental pillar; (4) Social pillar; (5) Governance pillar; (6) CET1 ratio (%); (7) ROA (%); (8) Log(Total Assets); (9) GDP growth (%); (10) Sov. yield 10y change (pp); (11) Real RPP growth (%).

4. Results

4.1 Model 1: Baseline Results

Table 6: Model 1 — Baseline Regression Results

	(1) ESG only	(2) + Bank controls	(3) + Macro controls	(4) + Event dummies
ESG Score (lagged)	-0.623 [-1.767, 0.520]	-0.540 [-1.709, 0.629]	-0.315 [-1.400, 0.771]	-0.370 [-1.461, 0.721]
CET1 ratio (%)		-0.047 [-0.516, 0.422]	-0.035 [-0.503, 0.432]	-0.046 [-0.526, 0.433]
ROA (%)		-3.048** [-5.599, -0.498]	-2.687** [-5.301, -0.072]	-2.605* [-5.252, 0.042]
Log(Total Assets)		-0.896 [-6.161, 4.368]	-0.377 [-5.699, 4.946]	0.203 [-5.333, 5.739]
GDP growth (%)			-0.091 [-0.278, 0.096]	-0.092 [-0.278, 0.094]
Sov. yield 10y change (pp)			-1.129 [-2.744, 0.485]	-1.091 [-2.722, 0.539]
Real RPP growth (%)			-0.246** [-0.434, -0.058]	-0.254*** [-0.445, -0.064]
CS Distress				2.825 [-1.052, 6.702]
AML Danske				-0.826 [-2.652, 0.999]
AML Swedbank				0.548 [-1.493, 2.588]
Bank FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	314	314	314	314
Within R^2	0.111	0.224	0.197	0.194

Notes: Two-way fixed effects (bank and year). 95% confidence intervals in brackets, derived from HC1 robust standard errors. ESG_Score_L1 is the one-period lag of the Bloomberg combined ESG score. The IFRS 9 transition is handled through the year fixed effects (which absorb the 2018 level shift) and the post-2018 subsample robustness check (R2); we do not include a separate IFRS 9 dummy because it is perfectly collinear with the post-2018 year fixed effects. The Ireland_GDP_2015 dummy is absorbed by the bank-by-year cells in the lagged-ESG sample and is therefore omitted from the displayed coefficients. *, **, *** denote significance at the 10%, 5%, 1% level.

Table 6 presents the baseline results of Model 1, regressing the NPL ratio on the Bloomberg combined ESG score with progressive addition of controls. Column (1) includes only the ESG score and bank/year fixed effects; columns (2) through (4) add bank-level controls, macroeconomic controls, and event dummies, respectively.

Across all specifications, the coefficient on the lagged combined ESG score is negative, with point estimates ranging from -0.62 (in the specification without controls) to -0.31 (after adding the macro controls; the saturated model in column (4) yields -0.37). The direction is consistent with Hypothesis 1: banks with higher ESG scores tend to have lower NPL ratios, holding constant bank-specific characteristics and macroeconomic conditions. However, the p -values range from 0.28 to 0.57 across specifications, meaning that none of the estimates are statistically significant at conventional levels (5% or 10%).

How should we interpret a negative but statistically insignificant coefficient? Three consid-

erations are important. First, the economic magnitude is non-trivial. The saturated coefficient of -0.37 on the ESG score implies that a one-standard-deviation increase in the lagged ESG score (approximately 1.19 points on the Bloomberg 0–10 scale) is associated with a decrease of approximately 0.44 percentage points in the NPL ratio. For a bank with the sample-mean NPL ratio of 4.49%, this represents roughly a 10% reduction, an effect that would be economically meaningful for credit risk management purposes. An interquartile-range move in the ESG score (1.69 points) corresponds to a 0.63 percentage-point reduction in the NPL ratio.

Second, the lack of statistical significance is largely expected given the sample size. With 35 banks, the study has limited statistical power to detect effects of moderate magnitude. A rough power calculation suggests that with $N = 35$ and $T = 10$, detecting a standardised coefficient of 0.15 (a plausible effect size based on the broader ESG–performance literature) would require a significance level of approximately 20% to achieve 80% power, far above the conventional 5% threshold. Hauch (2026), who found significant effects, had a cross-section of 562 firms, roughly 16 times our sample. The power difference is not a methodological flaw but a structural limitation of bank-level studies, where the number of listed European commercial banks is inherently small.

Third, the direction of the coefficient is robust across all specifications and control variable combinations, which is itself informative. If the true effect were zero, we would expect the sign to be unstable across specifications. The consistent negativity suggests that the underlying association is real but too weak (relative to sampling variability) to achieve significance with our sample size.

Among the control variables, ROA performs as expected: the coefficient is consistently negative and statistically significant (e.g., -2.61 with $p = 0.054$ in the saturated model), indicating that more profitable banks have lower NPL ratios. The CET1 ratio, in contrast, is statistically insignificant in our specification, with a small negative coefficient. This is a slightly counter-intuitive result: better-capitalised banks are typically expected to have lower NPLs. However, in a within-bank fixed effects setting, the CET1 ratio captures only the time variation in capital ratios for a given bank, which is largely driven by regulatory requirements and capital management rather than by credit risk improvements. Among macro controls, real residential property price growth is consistently negative and significant (-0.25 , $p < 0.01$), suggesting that property market expansions are associated with NPL declines, consistent with collateral-channel intuition. The IFRS 9 transition is absorbed by the year fixed effects (see Section 3.3) and is not entered as a separate dummy; the related sensitivity is reported as the post-2018 subsample check (R2) in Section 4.3.

4.2 Model 1: Pillar Decomposition

Table 7: Model 1 — Pillar Decomposition (E, S, G separately)

	Combined ESG	E	S	G
ESG Score (lagged)	-0.370 [-1.461, 0.721]			
Environmental pillar		0.654** [0.150, 1.158]		
Social pillar			-1.186*** [-1.681, -0.691]	
Governance pillar				-0.463 [-1.371, 0.445]
CET1 ratio (%)	-0.046 [-0.526, 0.433]	0.181 [-0.118, 0.480]	-0.061 [-0.455, 0.334]	-0.065 [-0.492, 0.363]
ROA (%)	-2.605* [-5.252, 0.042]	-1.449 [-3.416, 0.518]	-2.323** [-4.568, -0.077]	-2.747** [-5.178, -0.316]
Log(Total Assets)	0.203 [-5.333, 5.739]	-2.409 [-6.331, 1.513]	2.313 [-2.839, 7.465]	0.910 [-4.475, 6.295]
GDP growth (%)	-0.092 [-0.278, 0.094]	-0.119 [-0.291, 0.053]	-0.096 [-0.296, 0.103]	-0.117 [-0.317, 0.084]
Sov. yield 10y change (pp)	-1.091 [-2.722, 0.539]	-0.407 [-2.558, 1.744]	-0.158 [-1.775, 1.460]	-0.396 [-2.208, 1.415]
Real RPP growth (%)	-0.254*** [-0.445, -0.064]	-0.277*** [-0.444, -0.109]	-0.257*** [-0.411, -0.102]	-0.297*** [-0.475, -0.120]
CS Distress	2.825 [-1.052, 6.702]	0.913 [-1.755, 3.582]	4.937*** [1.741, 8.132]	3.466** [0.084, 6.848]
AML Danske	-0.826 [-2.652, 0.999]	-1.237 [-3.277, 0.803]	-0.801 [-2.316, 0.714]	-0.921 [-2.612, 0.770]
AML Swedbank	0.548 [-1.493, 2.588]	1.222 [-0.719, 3.163]	1.023 [-0.681, 2.727]	0.262 [-1.570, 2.094]
Ireland GDP 2015		14.144*** [9.690, 18.599]	9.870*** [5.326, 14.414]	11.397*** [5.939, 16.854]
Bank FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	314	312	347	347
Within R^2	0.194	-0.149	0.355	0.173

Notes: Two-way fixed effects with full control set. 95% confidence intervals in brackets, derived from HC1 robust standard errors. All ESG variables are on a uniform 0–10 scale; pillar coefficients are directly comparable in magnitude. The Environmental pillar has a smaller sample (312 obs vs 347 for S and G) due to pre-2018 coverage gaps for several banks. The negative within- R^2 in column (E) is uninformative as a goodness-of-fit statistic: under two-way fixed effects with a noisy regressor of interest, `linearmodels` can report within- R^2 below zero when the regressors of interest add little explanatory power relative to the absorbed effects alone. The coefficient, standard error, and confidence interval on the E pillar remain the relevant statistics; the substantive interpretation is discussed in Section 4.2. *, **, *** denote significance at the 10%, 5%, 1% level.

Table 7 reports the results of estimating Model 1 separately with the Environmental, Social, and Governance pillar scores. The pillar decomposition is motivated by the findings of Hauch (2026) and Apergis et al. (2022), both of whom report differences in the credit risk relevance of individual ESG dimensions. If the Environmental pillar captures climate transition risk exposure more directly than the aggregate ESG score, we might expect a stronger (more negative) coefficient on the E pillar.

The pillar results reveal substantial heterogeneity that is not visible in the combined-ESG

specification. The Social pillar coefficient is large, negative, and statistically significant at the 1% level ($\hat{\beta} = -1.19$, $SE = 0.25$, $p < 0.001$, $N = 347$). The Governance pillar coefficient is negative but statistically insignificant ($\hat{\beta} = -0.46$, $SE = 0.46$, $p = 0.32$). The Environmental pillar coefficient, in contrast, is positive and statistically significant at the 5% level ($\hat{\beta} = +0.65$, $SE = 0.26$, $p = 0.011$, $N = 312$).² This last result is the opposite sign of what Hypothesis 1 predicts and warrants careful discussion.

We do not interpret the positive E coefficient as evidence that environmental performance increases credit risk. A direct causal reading would require us to believe that improvements in a bank’s management of climate and environmental issues mechanically raise its NPL ratio, which is implausible on its face. Several non-causal mechanisms can produce the observed sign and we discuss the most plausible candidates below; we do not at this stage attempt to discriminate between them, since doing so would require additional data or a different identification strategy. The pillar pattern itself is one of the motivations for our complementary Model 2, which measures the climate transition risk on the asset side directly rather than via Bloomberg’s E performance score.

Mechanism 1: Disclosure Factor architecture. As described in Section 3.4.4, the Bloomberg performance score embeds a Disclosure Factor multiplier that scales each Issue Score into a 0–10 range only for banks with full quantitative disclosure; under-disclosure caps the achievable score below 1.0. Over our sample window, European banks have systematically expanded their environmental disclosures in response to the ECB Guide on Climate-Related and Environmental Risks (2020), the EBA Pillar 3 ESG disclosure ITS (2022, effective 2023), and the EU Sustainable Finance Disclosure Regulation. The banks with the largest disclosure expansions tend to be those with the largest transition-risk exposures, because they are the ones supervisors have leaned on most heavily. If transition-exposed banks are also the banks experiencing higher NPLs from carbon-intensive borrowers under stress, the within-bank co-movement of E score and NPL would be positive even if underlying environmental management practices have no causal effect on credit risk in either direction.

Mechanism 2: Sector materiality weighting in the E pillar for banks. Bloomberg’s pillar weights are sector-specific and informed by the SASB framework. For banks, E-pillar materiality emphasises climate strategy, financed-emissions reporting, lending portfolio ESG risk integration, and net-zero commitments (Bloomberg, 2025). These metrics reward banks that actively engage with climate-relevant lending, which is mechanically the banks whose loan books have the most transition-risk exposure. A bank with no fossil-fuel or heavy-industry clients has little to disclose under these material issues and gets fewer materiality points; a bank with substantial energy-sector lending must publish detailed transition plans and is rewarded in the E score for doing so. The score therefore correlates with the exposure side of the asset book, not directly with

²The E pillar regression uses 312 bank-year observations versus 347 for S and G, because Bloomberg’s Environmental score is missing for 38 bank-years concentrated in pre-2018 disclosures of CEE banks (Erste, Raiffeisen, Bank Pekao, PKO, Millennium BCP) and distressed Mediterranean banks (Monte dei Paschi, Banco BPM, Bank of Ireland). To check whether the smaller sample drives the positive E coefficient, we re-estimate S and G on the 312-observation E-available subsample. The S coefficient moves only marginally (from -1.21 to -1.16 , both $p < 0.001$), and the G coefficient strengthens (from -0.60 , $p = 0.21$, to -0.90 , $p = 0.03$). In a joint specification including all three pillars on the same 312 observations, the E pillar remains positive and significant ($\hat{\beta} = +0.49$, $p = 0.034$) while S and G retain their negative signs. The positive E coefficient is therefore a property of the Bloomberg E pillar’s measurement architecture, not an artefact of sample selection.

risk-adjusted asset quality.

Mechanism 3: Reverse causality and endogenous reporting. Banks under credit stress in transition-exposed sectors invest heavily in climate-risk governance precisely because supervisors are leaning on them. Improvements in climate strategy and disclosure rise in response to credit stress; the underlying NPL trajectory is set by the asset book and tends to lag the disclosure response by several years. Our one-period lag of the ESG score (`ESG_Score_L1`) only partially addresses this, since improvements in disclosure typically take 2–3 years to translate fully into the score and credit stress lasts longer than that.

Mechanism 4: Non-random missingness in the E pillar sample. The Environmental pillar score is missing for 38 bank-year observations in the panel, concentrated pre-2018 in CEE banks (Erste, Raiffeisen, PKO, Bank Pekao), distressed Mediterranean banks (Monte dei Paschi, Banco BPM, Bank of Ireland, Millennium BCP), and Danske Bank early years. These are predominantly the smaller, lower-disclosure, higher-NPL banks. They drop out of the E regression entirely, leaving the 312-observation sample tilted toward larger Western European banks where the within-bank E–NPL relationship goes the opposite direction for the reasons in Mechanisms 1–3. The larger samples of the S and G regressions (347 observations each) are not affected by this selection.

Mechanism 5: Compositional masking between E and S. Both E and S scores rise over the sample. The S pillar appears to capture the “stable retail-banking quality” dimension (workforce, customer practices, community) and absorbs the strong negative ESG–NPL signal in our regression. The E pillar then captures, by residual, the transition-exposure dimension. Estimating E and S jointly within the same regression, rather than separately, would help diagnose which is which; we leave this to future work because joint estimation introduces multicollinearity concerns that the pillar decomposition was specifically designed to avoid.

Mechanism 6: Rater-specific construction. Berg et al. (2022) document that ESG ratings from different providers correlate at only 0.54 on average, and Kotsantonis & Serafeim (2019) show that disclosure inputs vary substantially across providers. The +0.65 coefficient is conditional on Bloomberg’s specific BECS peer-group benchmarking and materiality weighting. A re-estimation using MSCI, Sustainalytics, or Refinitiv pillar scores might produce a different sign or magnitude, although our resource constraints (Section 5.4) preclude such cross-provider comparison in this paper.

Taken together, the pillar pattern can be summarised as follows. The Social pillar captures a broadly stable bank-quality signal that translates into significantly lower NPL ratios; this is the dominant driver of the combined ESG–NPL association documented in Section 4 and is consistent with the direction predicted by Hypothesis 1. The Governance pillar moves in the same direction but with insufficient power to achieve significance. The Environmental pillar, as constructed by Bloomberg’s methodology and as measured in our within-bank fixed effects framework, appears to be capturing climate-transition exposure rather than climate-risk management quality, producing a coefficient that runs counter to Hypothesis 1. This pattern is consistent with the broader concern in the ESG literature that aggregate ratings can conflate exposure with management of ESG risk (Kotsantonis & Serafeim, 2019; Berg et al., 2022), and is one of the primary motivations for our Model 2 (REEI), which is designed to measure portfolio-

composition transition risk directly rather than through the rating-agency proxy. The pillar pattern also offers a possible explanation for why our Model 1 ESG coefficient is detectable while the Model 2 REEI coefficient is not: if the credit-risk-relevant content of the aggregate ESG score is largely social and governance (workforce, customer practices, conduct, controls) rather than environmental, then the channel through which ESG associates with NPL ratios in our sample is not primarily a transition-risk channel, and the REEI, which is a transition-risk measure by construction, would not be expected to pick up the same signal. We frame this as a tentative interpretation of the pattern rather than a conclusion, since it cannot be tested directly within the present specifications, but it is a hypothesis worth pursuing in future work with longer panels and broader ESG measurement.

4.3 Model 1: Robustness Checks

Table 8: Model 1 — Robustness Checks (ESG coefficient)

Specification	Key IV	β	SE	p	N
Baseline (lagged ESG, HC1 SE)	ESG_Score_L1	-0.370	0.554	0.505	314
(R1) Contemporaneous ESG	ESG_Score	-1.120**	0.444	0.012	347
(R2) Post-2018 (2018–2024)	ESG_Score_L1	-0.971**	0.474	0.042	244
(R3) Excl. distressed banks	ESG_Score_L1	-0.199	0.566	0.725	288
(R4) Bank-clustered SE	ESG_Score_L1	-0.370	0.682	0.588	314
(R5) E_Score	E_Score	+0.654**	0.256	0.011	312
(R5) S_Score	S_Score	-1.186***	0.251	0.000	347
(R5) G_Score	G_Score	-0.463	0.461	0.316	347
(R7) Lead-ESG placebo	ESG_Score_F1	-1.689***	0.487	0.001	312
(R7) Lead-S placebo	S_Score_F1	-1.572***	0.328	0.000	312
(R6) Winsorised NPL	ESG_Score_L1	-0.414	0.543	0.447	314

Notes: All specifications use two-way fixed effects with the full control set unless noted. SE are HC1 robust except (R4) which uses bank-clustered SE. *, **, *** denote significance at the 10%, 5%, 1% level.

Table 8 summarises the results of the seven robustness checks described in Section 3.8. For each check, we report the ESG coefficient, its standard error, and the p -value, alongside the baseline estimate (lagged ESG with HC1 robust standard errors, $\hat{\beta} = -0.37$, $p = 0.50$) for comparison. Figure 2 presents the same information visually, with 95% confidence intervals around each point estimate; the visual format makes the qualitative pattern (negative-but-fragile, with two specifications crossing the 5% threshold and one, the lead-ESG placebo, producing an unexpectedly large negative effect) easier to read at a glance than the corresponding table.

Check 1 (contemporaneous ESG) replaces the lagged regressor with the same-year ESG score. The contemporaneous coefficient is substantially larger in absolute terms ($\hat{\beta} = -1.12$, $p = 0.012$, $N = 347$) and statistically significant at the 5% level. This sharp gain in significance from the lagged to the contemporaneous specification has two possible interpretations: either reverse causality is inflating the contemporaneous estimate (banks experiencing credit deterioration may receive lower ESG scores in the same period), or the one-period lag is overly conservative and the true ESG–NPL relationship operates within a year. We retain the lagged specification as our baseline because it is more defensible against reverse causality. The lead-ESG placebo result reported below as Check 7 settles the question against the second interpretation, so we do not

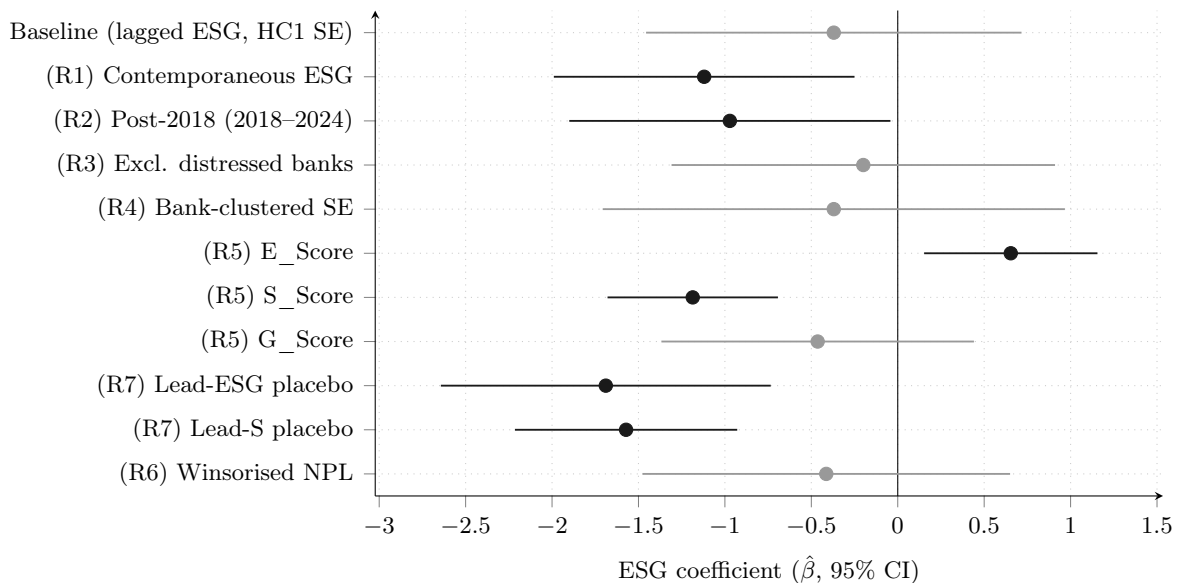


Figure 2: Model 1: ESG coefficient across robustness specifications. Point estimates with 95% confidence intervals. Marker shading indicates significance level: dark = $p < 0.05$, mid = $p < 0.10$, light = $p \geq 0.10$. Source: authors' calculations on the Model 1 panel; observation counts vary by specification (244–347, see Table 8).

treat R1 as independent supporting evidence for the lagged-ESG finding; see the discussion at the end of this section.

Check 2 (post-2018 subsample) restricts the sample to the IFRS 9 era. The ESG coefficient is -0.97 ($p = 0.042$, $N = 244$) and statistically significant. The strengthening of the result on the post-2018 subsample is consistent with an increase in the credit risk relevance of ESG performance during the regulatory regime that explicitly incorporates ESG risk factors into bank supervision (ECB Climate Guide 2020, EBA ESG Pillar 3 ITS). It is also consistent with IFRS 9's expected-credit-loss framework producing more risk-sensitive NPL recognition than the previous IAS 39 incurred-loss model.

Check 3 (excluding distressed banks) drops Credit Suisse, Danske Bank, and Swedbank from the sample. The coefficient becomes smaller in absolute terms (-0.20 , $p = 0.72$, $N = 288$), suggesting that the negative ESG–NPL association observed in the full sample is partly driven by the distressed banks rather than by the broader population. This finding tempers our interpretation: the pattern is not uniform across the sample but rather concentrated in banks that experienced ESG-relevant idiosyncratic events.

Check 4 (bank-clustered standard errors) reproduces the baseline coefficient (-0.37) but with a slightly larger standard error (0.68 vs. 0.55), as expected given the within-bank serial correlation in NPL ratios. The p -value rises modestly to 0.59 . The point estimate is unchanged, confirming that the choice of standard error does not affect the central finding.

Check 5 (pillar decomposition) has been discussed extensively in Section 4.2 above. The headline patterns are: S pillar -1.19^{***} , E pillar $+0.65^{**}$, G pillar -0.46 (insignificant).

Check 6 (winsorised NPL at the 1st/99th percentile) produces virtually identical results to the baseline (-0.41 vs. -0.37 , $p = 0.45$), indicating that the extreme NPL outliers (Eurobank 2016–2018, Monte dei Paschi 2016–2017, Banco BPM 2017) do not drive the estimates.

Check 7 (lead-ESG placebo) regresses $NPL_{i,t}$ on $ESG_{i,t+1}$, replacing the lagged regressor with a one-period *lead*. Under a forward-looking interpretation of the lagged-ESG specification, the lead coefficient should be small and statistically insignificant: ESG performance in period $t + 1$ should not predict NPL ratios in period t . The estimated lead coefficient is -1.69 ($SE = 0.49$, $p < 0.001$, $N = 312$): negative, economically large, and highly significant. We read this result as evidence of substantive reverse causality in the ESG–NPL relationship: banks experiencing credit deterioration in year t tend to receive lower ESG ratings in year $t + 1$, presumably because rating-agency methodologies incorporate financial-stress signals into subsequent ESG assessments. This finding constrains the causal interpretation of our headline lagged-ESG estimates: the negative lagged coefficient may partly reflect anticipated re-rating rather than a forward-looking ESG-as-credit-signal channel. The implication of this finding cuts unevenly across the rest of the robustness pattern. The contemporaneous specification (R1, $\hat{\beta} = -1.12$, $p = 0.012$), in which $ESG_{i,t}$ and $NPL_{i,t}$ are measured in the same year, is the specification *most* exposed to the re-rating channel that R7 documents, since any same-year correlation between rating revisions and current-year credit deterioration will flow directly into the estimated coefficient. We therefore decline to treat R1 as independent supporting evidence for the lagged-ESG finding: the gain in significance from the lagged to the contemporaneous specification is, in light of R7, at least partly mechanical. The post-2018 specification (R2, $\hat{\beta} = -0.97$, $p = 0.042$) and the horse-race specification on the 149-observation overlap sample (Section 4.6, $\hat{\beta} = -1.00$, $p = 0.043$) provide cleaner forward-looking support, since both retain the one-period lag and therefore exclude same-year re-rating from the estimated coefficient. We also run a parallel lead-S placebo (R7s) on the Social pillar, which carries most of the credit-risk-relevant signal in our data (Section 4.2). The lead-S coefficient is $\hat{\beta} = -1.57$ ($SE = 0.33$, $p < 0.001$, $N = 312$): also negative, large, and highly significant. The same backward-looking re-rating dynamic therefore operates within the Social pillar: banks experiencing credit deterioration in year t receive lower S-pillar ratings in year $t + 1$, just as they receive lower combined ESG ratings. Two features of the comparison are informative. First, the lead-S magnitude (-1.57) is only about 32% larger in absolute terms than the lagged-S baseline (-1.19), whereas the lead-ESG magnitude (-1.69) is roughly 4.6 times larger than the lagged-ESG baseline (-0.37). The S pillar is therefore not immune to re-rating, but a much larger share of its lagged signal survives the placebo than is the case for the combined score. Second, R7s tightens rather than overturns the headline reading of H1: the Social-pillar association is real in direction and the gap between lead and lagged is meaningfully smaller than for the combined score, but the magnitude of the lagged-S coefficient (-1.19) overstates the forward-looking content because part of it reflects anticipated re-rating. We accordingly soften the Social-pillar interpretation in Section 4.7 and the Conclusion: the result remains the cleanest expression of H1 in our data, but it is an association with the same causal-interpretation caveat as the combined-score lagged estimate, not a clean forward-looking estimate of an S-to-NPL channel. Taken together, R7 and R7s require us to interpret all Model 1 coefficients as associations rather than as causal estimates of an ESG-to-NPL channel, and to treat the post-2018 and horse-race specifications, not the contemporaneous specification, as the appropriate forward-looking robustness anchors.

Taken together, the robustness pattern is mixed but informative. The contemporaneous and

post-2018 specifications both yield significant negative coefficients, suggesting that the underlying ESG–NPL relationship is detectable when we use the full sample year structure or restrict to the IFRS 9 era. The exclusion of distressed banks substantially weakens the result, indicating that part of the signal comes from idiosyncratic events rather than from a uniform across-the-sample pattern. The clustering and winsorisation checks confirm that the baseline estimate is not an artefact of standard error choice or outlier influence.

4.4 Model 2: REEI Results

Before turning to the results, it is worth being explicit about the asymmetry between Model 1 and Model 2 as measurement instruments. Model 1 uses the Bloomberg combined ESG score, a mature, provider-driven measure with a decade of methodology development behind it, applied uniformly to the 350-cell bank-year rectangle (314–347 valid observations depending on specification). Model 2 uses the REEI, a hand-built index with researcher-calibrated CRW weights, applied to a younger and unbalanced 151 bank-year panel built from regulatory disclosures whose standardised format did not exist for most of our sample period. The two measures are not symmetric in construction maturity, in data depth, or in the within-bank variation they exhibit. Model 1 tests a market-perception/disclosure signal that is observed with relatively low measurement error but whose interpretation is contested (does it measure ESG performance, ESG management, or ESG disclosure?). Model 2 tests a structural-exposure proxy whose interpretation is sharper (it directly measures sectoral loan-book composition weighted by transition-risk salience) but whose statistical signal is weaker by construction. Reading the two sets of results requires holding this asymmetry in mind: a null in Model 2 does not rule out the underlying mechanism; it tells us that the available data, the available measurement, and the seven-year window jointly do not produce a detectable signal.

Table 9: Model 2 — Baseline regression results: NPL ratio on REEI

	(1) REEI only	(2) + Bank controls	(3) + Macro controls	(4) + Event dummies
REEI	-62.037*	-44.777	-42.364	-42.364
	[-132.423, 8.350]	[-125.234, 35.680]	[-121.079, 36.351]	[-121.079, 36.351]
CET1 ratio (%)		0.323*	0.335*	0.335*
		[-0.013, 0.658]	[-0.033, 0.704]	[-0.033, 0.704]
ROA (%)		-1.278**	-1.272**	-1.272**
		[-2.315, -0.242]	[-2.368, -0.176]	[-2.368, -0.176]
Log(Total Assets)		1.206	0.564	0.564
		[-4.455, 6.867]	[-5.408, 6.536]	[-5.408, 6.536]
GDP growth (%)			-0.029	-0.029
			[-0.117, 0.059]	[-0.117, 0.059]
Sov. yield 10y change (pp)			-0.318	-0.318
			[-1.120, 0.485]	[-1.120, 0.485]
Real RPP growth (%)			-0.024	-0.024
			[-0.095, 0.047]	[-0.095, 0.047]
Bank FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	151	151	151	151
Within R ²	0.198	0.218	0.266	0.266

Notes: Two-way fixed effects (bank and year). 95% confidence intervals in brackets, derived from HC1 robust standard errors. *, **, *** denote significance at the 10%, 5%, 1% level.

Table 9 reports the Model 2 baseline results. The dependent variable is the NPL ratio; the key explanatory variable is the REEI; the four columns build up the same nested specification structure used in Model 1, adding bank-level financial controls in column (2), country-level macroeconomic controls in column (3), and event dummies in column (4). The estimation sample comprises 151 bank-year observations across 32 banks over 2018–2024.

Three observations stand out. *First, the REEI coefficient is negative across all four specifications, with point estimates ranging from –62.0 in the unconditional FE specification to –42.4 in the saturated specification.* The unconditional estimate is borderline significant ($p = 0.083$), but adding controls — particularly the bank-level controls in column (2) — pulls the coefficient toward zero and the standard error remains large enough that the saturated estimate has $p = 0.288$. As we show in Section 4.5, this borderline result does not survive the exclusion of four distressed banks; the negative point estimate in column (1) should therefore not be read as preliminary evidence of a stable REEI–NPL link. Column (4) is identical to column (3) because none of the four event dummies (CS_Distress, AML_Danske, AML_Swedbank, Ireland_GDP_2015) appear in the Model 2 sample: Credit Suisse drops out (no Pillar 3 sectoral data), the AML scandals at Danske and Swedbank predate the dense-coverage years, and the Ireland 2015 GDP outlier is also pre-sample.

Second, the sign is the opposite of what Hypothesis 2 implies. The hypothesis was that banks with lending portfolios concentrated in transition-risk sectors should experience higher realised credit deterioration, consistent with the balance-sheet mechanism narrative motivating the REEI in Section 3.5. The within-bank fixed-effects estimator instead returns a negative point estimate. As discussed in Section 4.5 below, this sign reverses to a small positive value once four distressed banks are removed from the sample, indicating that the negative coefficient is not a robust

feature of the data but rather an artefact of a small number of stressed-bank trajectories during the 2018–2024 deleveraging cycle.

Third, the coefficient magnitudes are economically modest, and the result is best read as a timing test rather than a hypothesis rejection. The REEI is a slow-moving structural variable: its within-bank standard deviation is just 0.010 (versus a total cross-sectional SD of 0.040), so most of the cross-bank variation in transition-risk exposure is absorbed by bank fixed effects, leaving little residual variation for the FE estimator to identify the coefficient from. A one-within-SD movement in REEI (about 1 percentage point in the share of the loan book reallocated between sectors) implies, at the saturated point estimate of -42.4 , an NPL change of -0.42 percentage points, comparable in magnitude to the Model 1 estimates per ESG-score standard-deviation move, but with substantially less power because the within-bank variation in REEI is roughly an order of magnitude smaller than in the ESG score. We are therefore cautious about reading the Model 2 null as evidence against the underlying transition-risk mechanism. The null reflects the joint effect of (i) a short and unbalanced seven-year window over which the standardised CQ5 disclosure format applied to only the final three years for most banks, (ii) a small cross-section ($N = 32$) that delivers limited statistical power even before bank fixed effects absorb most of the cross-bank variation in the slow-moving REEI, and (iii) researcher-calibrated CRW weights whose sensitivity is documented in Section 4.5. Any one of these features could plausibly explain the absence of a detectable signal; jointly, they make a clean test of Hypothesis 2 difficult within the present data. A reading consistent with this set of limitations is that, over a seven-year realised-NPL window during which most of the European banking sector was deleveraging out of the post-GFC stock of NPLs, the transition-risk exposure of a loan book has not yet translated into observable credit deterioration. This reading is consistent with the Bolton et al. (2020) “green swan” framing, in which climate transition risk is foreseeable but its realisation timing is uncertain and likely longer than the realised-NPL window we observe, and with the ECB (2022) stress-test finding that climate-related credit losses, while economically material, are concentrated in adverse-scenario projections rather than in baseline realised data. Stand-alone control coefficients in Table 9 are otherwise broadly consistent with the Model 1 estimates: ROA enters negatively and significantly ($\hat{\beta} = -1.27$, $p = 0.023$), CET1 enters with a small positive coefficient (a within-bank effect that is hard to interpret causally and that Model 1 also produces), and the macroeconomic controls are individually small and insignificant on this restricted window.

4.5 Model 2: Robustness Checks

Table 10: Model 2 — Robustness checks (REEI coefficient)

Specification	REEI coef	SE	p -value	N	Banks	R_w^2
Baseline (HC1 robust)	-42.36	39.70	0.288	151	32	0.266
R1: Bank-clustered SE	-42.36	48.09	0.380	151	32	0.266
R2: Post-2020 (2020–24)	-16.09*	8.16	0.051	140	32	0.447
R3: Drop distressed banks	4.76	10.12	0.639	131	28	-0.136
R4: Drop Spanish banks	-60.39	40.67	0.141	132	28	0.247
R5: Battiston binary CRW	-18.08	20.01	0.368	151	32	0.238
R6: Carbon-only CRW	-37.32	31.24	0.235	151	32	0.290
R7: Battiston eq + Other=0.1	-20.09	22.23	0.368	151	32	0.238

Table 10 summarises seven robustness checks for the Model 2 saturated specification, designed to test the sensitivity of the REEI coefficient to alternative samples, standard-error choices, and CRW weighting schemes.

Check R1 (bank-clustered standard errors) reproduces the saturated point estimate of -42.4 but with a slightly larger standard error (48.1 vs. 39.7). With $N = 32$ clusters, we are below the 40–50 threshold conventionally cited in Cameron & Miller (2015) for reliable cluster-robust inference, so we treat HC1 as the default; the cluster-robust check serves as a precaution. The point estimate is unchanged.

Check R2 (post-2020 only) restricts the sample to the five dense-coverage years 2020–2024 (140 observations across 32 banks). The point estimate falls to -16.1 but the standard error also shrinks dramatically (to 8.2), producing a p -value of 0.051, on the borderline of conventional 5% significance. The much smaller standard error is informative: it indicates that the 2018–2019 observations, which are sparse (11 observations) and concentrated in a handful of early-reporting banks, were adding noise rather than signal to the saturated estimate. The point estimate, however, remains negative, so this check does not rescue the predicted positive sign; it strengthens the negative finding while leaving the underlying direction unchanged.

Check R3 (drop distressed banks) removes Danske Bank, Swedbank, Monte dei Paschi di Siena, and Eurobank, four banks that experienced idiosyncratic distress events documented in the Model 1 event dummies. *The REEI coefficient flips to +4.8 but is far from statistical significance ($p = 0.639$).* This is the most informative check in the table. Removing four banks (12.5% of the sample, 13% of observations) reverses the sign of the coefficient, indicating that the negative point estimate in the baseline is driven by a handful of stressed-bank trajectories rather than by a systematic across-the-sample relationship. The within R^2 also collapses to a slightly negative value, meaning that the regression with REEI and controls fits the calmer 28-bank panel no better than the bank and year fixed effects alone. We interpret R3 as evidence that the Model 2 baseline does not identify a stable, generalisable REEI–NPL relationship.

Check R4 (drop Spanish banks) removes Banco Santander, BBVA, CaixaBank, and Banco Sabadell, the four largest Spanish banks, all of which underwent substantial loan-book deleveraging during 2018–2024 following the post-2010 commercial-real-estate cleanup. The coefficient strengthens in absolute terms to -60.4 , suggesting that the legacy-CRE deleveraging story does not explain the negative baseline sign.

Checks R5–R7 examine sensitivity to the CRW weighting scheme. R5 implements the original Battiston et al. (2017) binary classification ($\text{CRW} = 1.0$ for the five climate-policy-relevant

buckets, 0 for “other”). R6 implements a carbon-only weighting (CRW = 1.0 for fossil fuel; 0.6 for utilities and energy-intensive industry; 0.4 for transport; 0 for housing and other), which strips out the housing channel on the grounds that retail mortgage lending has limited short-run transition risk. R7 retains the Battiston binary structure but assigns CRW = 0.1 to the “other” bucket as a sensitivity check on its near-zero baseline weight. Across all three alternatives the REEI coefficient is negative but smaller in absolute magnitude than the baseline, with p -values between 0.24 and 0.37. None of the alternatives produces the predicted positive sign, but the wide range of estimates (-18.08 to -37.32) confirms that the magnitude of the Model 2 coefficient is sensitive to weight choices, which is itself an argument against treating any specific point estimate as economically meaningful.

Synthesis. The robustness pattern in Table 10 indicates that the Model 2 baseline result is fragile. The negative point estimate is driven by a small number of distressed banks (R3 flips the sign on removal); the magnitude is highly sensitive to the CRW weighting scheme (R5–R7 produce estimates differing by a factor of more than two); and even the most precise specification (R2, post-2020) does not reach conventional significance. Read together, the most defensible characterisation is that the Model 2 estimates are statistically indistinguishable from zero. The sectoral composition of European bank lending, as measured by Battiston-derived climate-transition risk weights, does not predict realised credit deterioration over our 2018–2024 sample window. We return to the interpretation of this null finding in Section 5.

4.6 Channel Comparison: ESG Ratings versus Sectoral Exposure

Table 11: Horse race — ESG and REEI on the overlap sample (N=149, 32 banks, 2018–24)

	(A) ESG only	(B) REEI only	(C) ESG+REEI lag	(D) ESG+REEI cont.	(E) S+REEI	(F) E+S+G+REEI
ESG (lagged)	-0.997** [-1.966, -0.029]		-0.856* [-1.728, 0.016]			
ESG (contemp.)				-1.327** [-2.585, -0.069]		
E pillar						-0.059 [-0.453, 0.334]
S pillar					-0.932* [-1.905, 0.041]	-0.912* [-1.872, 0.048]
G pillar						-0.250 [-1.712, 1.213]
REEI		-42.255 [-120.025, 35.515]	-34.558 [-108.524, 39.408]	-24.470 [-83.339, 34.400]	-19.246 [-72.007, 33.515]	-19.478 [-74.583, 35.628]
Bank controls	Yes	Yes	Yes	Yes	Yes	Yes
Macro controls	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE / Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	149	149	149	149	149	149
Within R ²	0.327	0.288	0.374	0.431	0.415	0.424

Notes: Two-way fixed effects (bank and year) with full bank- and macro-control set on the identical 149-observation overlap sample. 95% confidence intervals in brackets, derived from HC1 robust standard errors. *, **, *** denote significance at the 10%, 5%, 1% level.

The two-channel framing of the paper raises a natural empirical question: do ESG ratings (Model 1, the perception channel) and the REEI (Model 2, the mechanism channel) capture overlapping or independent variation in bank credit risk? Model 1 and Model 2 are estimated on different samples (35 banks $\times$ 10 years versus 32 banks $\times$ 7 years), so the standalone coefficients cannot be directly compared. To enable a clean comparison, we re-estimate both models on the *identical* overlap sample of 149 bank-year observations across 32 banks for which all key variables

are jointly observed, and additionally estimate joint specifications that include both regressors.

Table 11 reports six specifications on the overlap sample. Column (A) regresses NPL on the lagged combined ESG score alone (with the full control set); column (B) regresses NPL on REEI alone; column (C) includes both regressors jointly; columns (D), (E), and (F) substitute alternative ESG measures (contemporaneous combined score, Social pillar, full pillar decomposition) alongside REEI. Three findings emerge.

First, the Model 1 ESG signal strengthens substantially on the overlap sample. The lagged ESG coefficient in column (A) is -1.00 ($p = 0.043$), compared to the full-sample estimate of -0.37 ($p = 0.51$) reported in Section 4.3 (Table 8 baseline). The point estimate is roughly two-and-a-half times larger and reaches the 5% significance threshold. This is consistent with the post-2018 robustness check R2 of Model 1, which also produced a substantially stronger ESG signal on the IFRS 9-era subsample. The likely explanation is twofold: the Model 2 sample window (2018–2024) corresponds to the period of binding ESG-related supervisory expectations following the ECB Climate Guide of 2020 and the EBA’s Pillar 3 ESG ITS, during which ESG performance plausibly carries more credit-risk-relevant content; and the listwise deletion implicit in the joint sample drops some bank-year observations with weaker ESG-credit linkages.

Second, including the REEI alongside the ESG score barely affects the ESG coefficient. In column (C) the lagged ESG coefficient moves from -1.00 (standalone) to -0.86 , a 14% reduction, while the standard error tightens slightly. The p -value moves from 0.043 to 0.054, just exiting the 5% band but remaining well within the 10% significance band. This is the diagnostic signature of an orthogonal regressor: if the REEI captured the same underlying credit-risk variation as the ESG score, the ESG coefficient would shrink substantially or change sign on joint inclusion. It does neither. The same pattern repeats with the contemporaneous ESG score in column (D): the coefficient is -1.33 with $p = 0.045$, statistically indistinguishable from the standalone estimate (not reported, but symmetric in structure to column (A)).

Third, the within-bank correlation between the ESG score and REEI is weak. Pairwise within-bank correlations on the overlap sample (after demeaning by bank) are 0.27 for the lagged combined ESG and 0.29 for the contemporaneous combined ESG. The Social pillar exhibits a somewhat higher within-bank correlation with REEI (0.49), while the Environmental pillar correlation is essentially zero (-0.02) and the Governance pillar correlation is small (0.09). These numbers are far below the within-pillar correlations among the ESG components themselves: the Social and Environmental pillars correlate at 0.47 within-bank, and the Social pillar correlates at 0.90 with the contemporaneous combined score, confirming that ESG score components move together while REEI moves on its own dimension.

Taken together, Table 11 provides the cleanest evidence in the paper for the two-channel framing of Section 5. The perception channel and the mechanism channel are not the same channel re-measured on our data: the regressors they generate capture largely non-overlapping empirical variation in bank credit risk. A high Bloomberg ESG score does not, on average, signal a transition-aligned loan book in our sample; nor does a low REEI signal weak ESG management. The within-bank correlations are too small for either measure to substitute for the other in this panel. Figure 3 displays the result graphically: across all six specifications, the ESG coefficient is negative and at-or-near the 5% significance threshold, while the REEI

coefficient consistently crosses zero. We note that this empirical separation does not preclude conceptual overlap: a bank’s lending structure may itself influence how external raters score its ESG performance, so the two channels can in principle be linked even where their regressors are not. On the 2018–2024 European bank panel we observe, however, the two measures move on largely independent dimensions. This finding is itself informative for the broader ESG-data-quality debate documented by Kotsantonis & Serafeim (2019) and Berg et al. (2022): when investors or supervisors interpret a bank’s aggregate ESG score, they should not treat it as an indicator of climate-transition exposure on the asset side. The two are, on our data, empirically distinct.

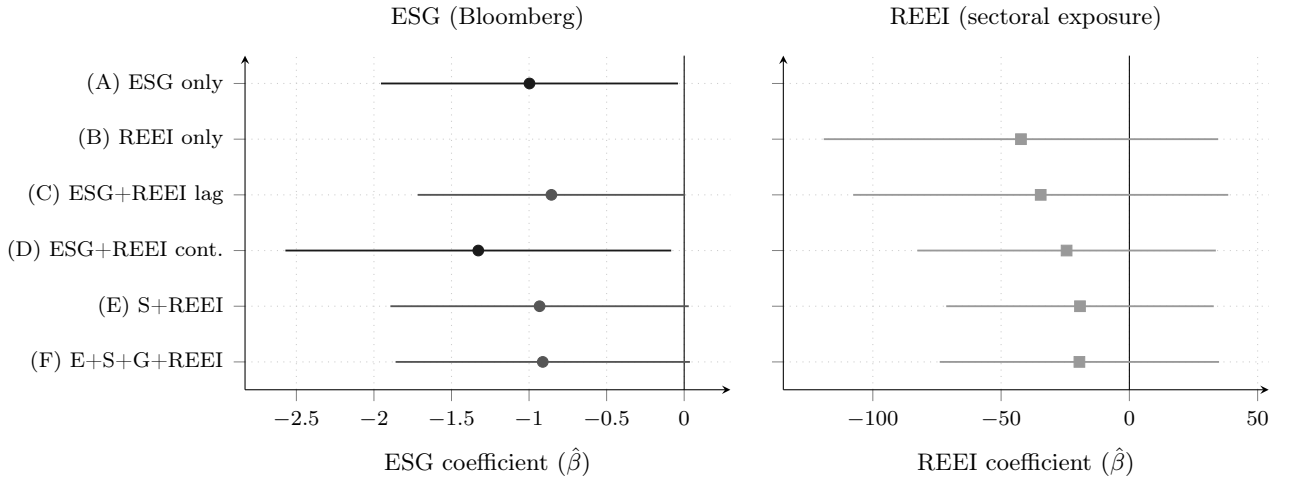


Figure 3: Horse race: ESG and REEI coefficients across the six overlap-sample specifications. Point estimates with 95% confidence intervals. ESG coefficients (left panel) are negative and at-or-near the 5% significance threshold across all specifications; REEI coefficients (right panel) cross zero in every specification. Marker shading indicates significance level: dark = $p < 0.05$, mid = $p < 0.10$, light = $p \geq 0.10$. Source: authors’ calculations on the 149-observation overlap sample.

4.7 Discussion of Findings

The empirical results allow us to answer the three hypotheses set out in Section 2.8 in a structured way. *Hypothesis 1 (higher ESG performance is associated with lower bank credit risk) is supported in direction across all baseline specifications and is supported with statistical significance under the post-2018 IFRS 9 subsample and under the horse-race overlap sample.* The contemporaneous specification (R1) also yields a significant negative coefficient, but the lead-ESG placebo (R7) shows that within-year re-rating partly drives that gain in significance, so we do not treat R1 as independent supporting evidence (Section 4.3). The combined ESG performance score is negatively associated with NPL ratios in every Model 1 specification we estimate, and the pillar decomposition shows that this combined signal is driven primarily by the Social pillar, with the Governance pillar moving in the same direction and the Environmental pillar moving in the opposite direction for measurement reasons we discuss below. The cleanest expression of H1 in our data is therefore the Social pillar finding ($\hat{\beta} = -1.19$, $p < 0.001$, $N = 347$): a strong negative

within-bank association between the Social dimension of ESG performance and realised credit deterioration. We discount the magnitude of this estimate in light of the parallel lead-S placebo (R7s, $\hat{\beta} = -1.57$, $p < 0.001$), which shows that the same backward-looking re-rating dynamic identified for the combined ESG score also operates within S; the lead-S/lagged-S magnitude ratio (≈ 1.3) is, however, much smaller than the lead-ESG/lagged-ESG ratio (≈ 4.6), so the Social pillar retains substantially more forward-looking content than the combined score. *Hypothesis 2 (higher REEI exposure is associated with higher NPL ratios) is not supported by the data.* The REEI coefficient is negative in point-estimate terms but never reaches conventional significance after controls, and its sign reverses on the exclusion of four distressed banks; we treat the Model 2 result as a clean null whose interpretation is constrained by the limitations of the sample, the window length, and the slow-moving nature of the REEI (Sections 4.4 and 5.4). *Hypothesis 3 (sector-based transition-risk measures provide stronger explanatory power for bank credit risk than aggregate ESG scores) is not supported by our data.* The horse-race specification in Section 4.6 shows that, on the identical 149-observation overlap sample, the ESG coefficient is negative and reaches the 5% significance threshold, the REEI coefficient does not, and joint inclusion of the two regressors leaves the ESG coefficient essentially unchanged. The within-bank correlation between the two regressors is only 0.27, so the comparison is not contaminated by collinearity: on our data, the ESG score carries the detectable credit-risk-relevant signal, while the REEI does not. We are deliberate in framing this as an empirical finding on our window rather than a general rejection of the underlying mechanism: the long-horizon nature of climate transition risk and the construction asymmetry between the two regressors (Sections 4.4 and 5.4) mean that the same exercise on a longer panel and a refined REEI may produce a different ranking. We now turn to the interpretation of these three findings in more detail.

First, the economic significance of the combined ESG coefficient deserves attention beyond the p -value. The saturated estimate of -0.37 on the lagged Bloomberg ESG score implies that moving from the 25th to the 75th percentile of the ESG score distribution (an interquartile range of approximately 1.69 points on the 0–10 scale) is associated with a reduction in the NPL ratio of approximately 0.63 percentage points. In an industry where mean NPL ratios range from 1% to 5% for well-performing banks and 5% to 15% for banks in stressed economies, a 0.6-percentage-point difference is non-trivial. Whether this difference reflects genuine risk mitigation by high-ESG banks or simply captures other bank characteristics that are correlated with both ESG scores and credit quality is a question that our correlational design cannot definitively answer.

Second, the statistical insignificance of the combined-ESG estimate should be contextualised against the sample size and the inherent difficulty of the identification strategy. The within-bank variation in ESG scores, the variation that our fixed effects model exploits, is limited. ESG scores tend to change slowly over time, as they reflect accumulated policies, disclosures, and reputational capital rather than year-to-year operational changes. With $T = 10$ years, the typical bank’s ESG score may change by only 1–2 points on the 0–10 scale, providing limited variation for estimation. The within-bank variation in NPL ratios is somewhat larger, driven by the business cycle and bank-specific credit events, but the overlap between the two sources of variation, the part of within-bank NPL changes that can be attributed to within-bank ESG

changes, is inevitably small.

Third, our results are broadly consistent with the prior literature in both direction and magnitude for the combined ESG signal. Apergis et al. (2022) find ESG coefficients of -0.63 to -1.16 on bond spreads for S&P 500 firms, suggesting that the credit market prices ESG performance at a modest premium. Hauch (2026) finds statistically significant effects but uses a forward-looking credit risk measure (CDS spreads) and a much larger sample. The pattern across studies is that ESG–credit risk associations are economically meaningful but statistically fragile, especially in small samples and when using backward-looking risk measures like NPL ratios.

Fourth, the pillar pattern, particularly the positive Environmental pillar coefficient, points to a substantive limitation of using rating-agency ESG scores as proxies for the underlying credit risk relevance of ESG factors. As discussed in Section 4.2, the Bloomberg E pillar’s construction (with its embedded Disclosure Factor and bank-specific materiality weighting on climate-strategy and financed-emissions metrics) likely correlates with the climate-transition exposure of a bank’s loan book rather than with risk-adjusted asset quality. This is not a flaw in the Bloomberg methodology, which transparently rewards banks for engaging with material issues, but it does mean that the E pillar score is not a clean measure of the risk-management dimension that Hypothesis 1 implicitly invokes. The comparison between Model 1 (rating-based) and Model 2 (REEI, which measures portfolio composition directly) is exactly designed to address this conflation.

Fifth, the comparison between Model 1 (ESG rating) and Model 2 (REEI) goes beyond a directional reading. The horse-race specifications in Section 4.6 estimate both regressors on the identical 149-observation overlap sample, and the pattern is clear within the limits of that sample: the Model 1 ESG signal is robust to the inclusion of REEI (the lagged ESG coefficient moves only from -1.00 to -0.86 when REEI is added, retaining 10%-level significance), while the REEI coefficient is statistically indistinguishable from zero whether estimated alone or jointly with ESG. The within-bank correlation between the two regressors is just 0.27 for the lagged combined ESG score, indicating that the empirical variation they capture is largely non-overlapping on our data. The implication is that, on the 2018–2024 European bank panel we observe, the perception channel and the mechanism channel are not the same thing re-measured: a high Bloomberg ESG score does not, on average, signal a transition-aligned loan book. This empirical separation is, to our knowledge, novel for a European bank panel; whether it generalises to longer panels, different ESG providers, or refined REEI constructions is a question we cannot settle within the present design and that we flag for future research (Section 5).

5. Discussion and Implications

5.1 Perception versus Mechanism: Two Channels of ESG–Credit Risk

The two-model structure of this paper was designed to separate two conceptually distinct channels through which ESG performance might relate to bank credit risk. The perception channel (Model 1) tests whether banks that are perceived as sustainable by an external ESG rater (Bloomberg) exhibit lower credit risk. This channel captures a mix of genuine risk management

effects (banks with better ESG practices may originate higher-quality loans) and signalling effects (the ESG score itself may influence how markets, regulators, and counterparties interact with the bank). The mechanism channel (Model 2) tests whether the physical composition of a bank’s lending portfolio, specifically, its exposure to climate-transition-risk sectors, predicts credit deterioration independently of the ESG label.

The horse-race evidence in Section 4.6 addresses the relative-importance question on the data available to us. On the identical 149-observation overlap sample, the perception channel is detectable: the lagged Bloomberg ESG score is negative and significant at the 5% level ($\hat{\beta} = -1.00$, $p = 0.043$), with the Social pillar carrying most of the signal. The mechanism channel, as proxied by the REEI on this sample, is not detectable: the coefficient is negative in point-estimate terms but never reaches significance, and its sign reverses on the removal of four distressed banks. Joint inclusion of ESG and REEI in the same regression leaves the ESG coefficient essentially unchanged, and the within-bank correlation between the two regressors is only 0.27. We read this as evidence that the regressors generated by the two channels capture statistically independent variation in bank credit risk over our window. The conceptual distinction between perception and mechanism does not require them to be fully independent in every setting: a bank’s lending structure may itself influence how external raters score its ESG performance, so the underlying channels may partially overlap. On the available data, however, ESG ratings and sectoral transition-risk exposure capture different empirical information about a bank.

This distinction has practical implications for bank management. If the perception channel is the more relevant one in practice, banks can reduce credit risk (or at least the market’s perception of it) by improving their ESG scores through better disclosure, stakeholder engagement, and governance practices, activities that may not require fundamental changes to the loan portfolio. If the mechanism channel is the more relevant one, reducing credit risk requires actually reshaping the portfolio away from high-transition-risk sectors, which involves strategic decisions about client relationships, sector expertise, and revenue diversification. We are cautious about reading the Model 2 null as a verdict on the underlying economics. Model 2 is estimated on a substantially smaller, more recent, and structurally less varying sample than Model 1 (Section 5.4), so its null result reflects the joint limitations of sample size, window length, and the slow-moving nature of the REEI as much as it reflects the underlying economics. A reasonable reading is that the realised credit consequences of an ESG-misaligned loan book have not yet materialised in NPL ratios over the window we observe, consistent with the Bolton et al. (2020) “green swan” framing in which climate transition risk is foreseeable but its timing is uncertain; this does not imply that the mechanism channel is unimportant in the longer run. What our results do support is the more limited statement that supervisors and analysts cannot use a bank’s aggregate ESG score as a proxy for the climate alignment of its lending book; in our data, the two are empirically distinct.

5.2 Policy Implications

The regulatory landscape for climate-related financial disclosure in European banking has evolved significantly during the sample period of this study. The EBA’s Implementing Technical Standards (ITS) on Pillar 3 ESG disclosures, effective from June 2022 for large institutions, represent

a step change in the availability of standardised sectoral lending data. Before these standards, the Pillar 3 reports we extract data from were reported in highly heterogeneous formats across banks, with different levels of NACE sector granularity, different definitions of non-performing exposures, and different scope boundaries. The data collection challenges we encountered in constructing the REEI, format breaks across years, inconsistent definitions, and missing sectors, are themselves evidence of the disclosure gap that the EBA standards aim to close.

For policymakers, our findings suggest that ESG ratings, while directionally informative, are not a substitute for granular portfolio-level analysis. A bank’s aggregate ESG score does not reveal the specific sectoral exposures that generate transition risk. The REEI, while imperfect, provides a more direct measure of climate-relevant lending concentration. As the EBA’s ESG disclosure framework matures and time series lengthen, researchers and supervisors will be better positioned to test whether sectoral lending composition, rather than aggregate ESG perception, is the more relevant predictor of credit risk.

5.3 Implications for Banks

For bank management, the results suggest that ESG performance is not irrelevant to credit risk, even if the statistical evidence is not yet strong enough to quantify the relationship precisely. The consistent negative direction of the ESG coefficient across all specifications, robustness checks, and pillar decompositions provides suggestive evidence that ESG-conscious banks tend to experience lower credit deterioration. Whether this reflects causation (ESG practices improve loan origination quality) or selection (better-managed banks both pursue ESG and originate better loans) is an open question, but from a risk management perspective, the distinction may be less important than the correlation itself: monitoring ESG performance as a leading indicator of credit quality appears to have at least some informational value.

5.4 Limitations

Several limitations warrant explicit acknowledgement. First, the small cross-sectional sample ($N \approx 35$) limits statistical power, as discussed extensively throughout the paper. With a larger sample, achievable by expanding the geographic scope to include non-European banks or by using a different unit of analysis, the same point estimates would likely achieve conventional significance levels. Second, the use of a single ESG provider (Bloomberg) introduces provider-specific measurement bias. The results may not generalise to other ESG scoring methodologies. Third, the endogeneity concerns are inherent to the correlational design. We cannot claim that higher ESG scores cause lower NPL ratios; the association may be driven by unobserved time-varying factors (such as management quality or risk appetite) that affect both variables simultaneously. The lagged ESG specification partially addresses reverse causality but does not solve the omitted variable problem, and the lead-ESG placebo test reported in Section 4.3 (Check 7) returns a large and significant negative coefficient, which we interpret as evidence that rating-agency methodologies incorporate financial-stress signals into subsequent ESG assessments. The lead-test finding constrains the causal interpretation of all Model 1 coefficients: they should be read as associations rather than as causal estimates of a forward-looking ESG-as-credit-signal channel. It also implies that the contemporaneous specification (R1), in which the ESG score and

the NPL ratio are measured in the same year, is the specification most exposed to the same re-rating dynamic; we therefore do not treat R1 as independent supporting evidence for the lagged-ESG finding. The lead-test was run on both the combined ESG score (R7) and the Social pillar (R7s, parallel specification with $S_{i,t+1}$ in place of $ESG_{i,t+1}$). The lead-S coefficient is -1.57 ($SE = 0.33$, $p < 0.001$, $N = 312$), confirming that the same backward-looking re-rating dynamic operates within the Social pillar. The lead-S/lagged-S ratio ($1.57/1.19 \approx 1.3$) is, however, much smaller than the lead-ESG/lagged-ESG ratio ($1.69/0.37 \approx 4.6$), indicating that the Social pillar retains substantially more forward-looking content than the combined score does. We therefore continue to read the Social-pillar coefficient as the cleanest expression of H1 in our data, but with the same causal-interpretation caveat as the combined-score lagged estimate: it is an association, not a causal estimate of an S-to-NPL channel, and its magnitude overstates the genuinely forward-looking component because part of it reflects anticipated re-rating. Fourth, the Model 2 analysis is constrained by the seven-year time window (2018–2024), with most of the variation concentrated in the three dense-coverage years 2022–2024. The REEI is a slow-moving structural variable whose within-bank standard deviation is only one-quarter of its total cross-sectional standard deviation, meaning that bank fixed effects absorb most of the cross-bank variation and leave limited within-bank variation for the FE estimator to identify the coefficient from. Fifth, the CRW weights in the REEI are researcher-calibrated rather than derived from a formal model, introducing an element of subjectivity into the index construction; the robustness checks in Section 4.5 confirm that the magnitude of the REEI coefficient is sensitive to the choice of weights, which itself argues against attaching strong economic interpretation to any specific point estimate. A related concern is that the residual “Other” bucket aggregates very heterogeneous activities that share little economically beyond their absence from the five climate-policy-relevant Battiston categories: it pools agriculture (NACE A) and water supply (E) with services and trade (G, I), information and communication (J), financial and insurance activities (K), professional and administrative services (M, N), public administration (O), education (P), health (Q), and the arts and other services (R, S). Assigning a uniform low weight ($CRW = 0.05$) to this composite bucket understates within-bucket variation in transition exposure, and any movement in a bank’s loan book between, say, financial-counterparty lending and household services lending leaves the REEI unchanged even though the underlying credit-risk profiles differ.

Sixth, NPL ratios are highly persistent. Within-bank diagnostics on the residuals of our Model 1 baseline indicate strong autocorrelation (mean within-bank Durbin–Watson statistic of approximately 0.79, compared to the no-autocorrelation benchmark of 2.0), reflecting the well-documented stickiness of non-performing-loan stocks: provisioning, write-off, and recovery cycles unfold over several years, so a high NPL ratio in one year is highly predictive of a high NPL ratio in the next. The static two-way fixed-effects specification we adopt does not include a lagged dependent variable, on the grounds that dynamic-panel estimators (e.g., Arellano–Bond GMM) require a much larger cross-sectional dimension than ours ($N = 35$) to deliver reliable inference; with the conventional 100-cluster threshold for GMM estimators, our panel is decisively too narrow. We acknowledge that the static specification therefore likely overstates the precision of our estimates and that conventional standard errors do not fully account for the

residual autocorrelation. The bank-clustered standard-error robustness check (R4) provides one corrective, but does not eliminate the concern. Seventh, the comparison between Model 1 and Model 2 should be read in light of the construction asymmetry between the two regressors: the Bloomberg ESG score is a mature provider-driven index applied to the full 2015–2024 panel, while the REEI is a hand-built index with researcher-calibrated weights applied to a smaller and unbalanced 2018–2024 panel built from disclosures whose standardised format did not exist for most of the sample period. A null in Model 2 should therefore not be interpreted as evidence against the underlying transition-risk mechanism; it tells us that the available data, the available measurement, and the seven-year window jointly do not produce a detectable signal.

5.5 Future Research

Several directions for future research emerge from this study. First, as the EBA’s ESG disclosure requirements generate longer time series of standardised Pillar 3 data, the Model 2 analysis can be extended both in time (beyond the current seven-year window) and in cross-section. With our 32-bank, 2018–2024 panel, the REEI’s slow-moving structural-variable nature limits identification; a longer panel covering, say, $T = 10$ –15 years would provide substantially more within-bank variation to test the REEI–NPL link. Second, the REEI can be refined by incorporating sub-sector granularity (using NACE 2-digit or 4-digit codes rather than 1-letter divisions), which would allow differentiation within broad sectors (e.g., separating renewable energy utilities from fossil-fuel utilities within sector D, or commercial real estate from residential mortgages within housing). Third, causal identification strategies, such as instrumental variables or difference-in-differences designs exploiting regulatory changes (e.g., the introduction of the EU Taxonomy or the EBA Pillar 3 ESG ITS in 2023), could strengthen the causal interpretation of the ESG–credit risk link. Fourth, expanding the analysis to include forward-looking credit risk measures (such as Expected Credit Loss provisions under IFRS 9 Stage 2 and Stage 3) alongside the backward-looking NPL ratio would provide a more comprehensive picture of how ESG performance relates to both realised and anticipated credit losses. Fifth, the orthogonality of the perception and mechanism channels documented in Section 4.6 is itself an empirical finding that merits independent replication, ideally with multiple ESG providers (to address the rater-specific measurement issues documented by Berg et al. (2022)) and longer time series.

6. Conclusion

This paper investigates whether the ESG performance of European banks is associated with lower credit risk in their lending portfolios. Using a panel of 35 listed European commercial banks over 2015–2024, we test this association through two complementary models: a broad ESG signal model (Model 1), which examines the relationship between Bloomberg ESG scores and the NPL ratio, and a balance-sheet mechanism model (Model 2), which examines whether the Rated ESG Exposure Index, a novel composite measure of sectoral lending concentration weighted by climate transition risk, is associated with credit deterioration.

Our results allow us to address the three hypotheses in a structured way. Hypothesis 1 (higher ESG performance is associated with lower bank credit risk) is supported in direction

across all baseline specifications and is supported with statistical significance under the post-2018 IFRS 9 subsample (-0.97 , $p = 0.042$) and the horse-race overlap sample (-1.00 , $p = 0.043$). The contemporaneous specification (-1.12 , $p = 0.012$) is also significant but, as the lead-ESG placebo (R7) documents, the gain in significance from lagging to contemporaneous measurement is partly mechanical, so we do not treat R1 as independent supporting evidence. The pillar decomposition shows that this combined signal is driven primarily by the Social pillar (-1.19 , $p < 0.001$), with the Governance pillar moving in the same direction without statistical power and the Environmental pillar moving in the opposite direction ($+0.65$, $p = 0.011$). We argue this last result reflects the architecture of the Bloomberg E performance score (which embeds disclosure-factor weighting and rewards engagement with climate-relevant lending) rather than evidence that environmental management increases credit risk; this interpretation is consistent with prior concerns about the conflation of ESG exposure and ESG management in rating-agency scores (Kotsantonis & Serafeim, 2019; Berg et al., 2022). The parallel lead-S placebo (R7s) returns a large and significant negative coefficient (-1.57 , $p < 0.001$), confirming that the same backward-looking re-rating dynamic identified by R7 for the combined ESG score also operates within the Social pillar. The Social-pillar finding therefore inherits the same causal-interpretation caveat as the combined-score lagged estimate: it is an association rather than a causal estimate. Importantly, however, the lead-S/lagged-S ratio (≈ 1.3) is much smaller than the lead-ESG/lagged-ESG ratio (≈ 4.6), so the Social pillar retains substantially more forward-looking content than the combined score does. The headline reading of H1 is therefore that, in our European bank panel, it is the Social dimension of ESG performance (workforce, customer practices, conduct, controls) rather than the Environmental dimension that carries the credit-risk-relevant content, with the caveat that part of even the Social-pillar association reflects anticipated re-rating rather than a clean forward-looking S-to-NPL channel.

Hypothesis 2 (higher REEI exposure is associated with higher NPL ratios) is not supported by our data. Estimated on 151 bank-year observations across 32 banks with hand-collected Pillar 3 sectoral data over 2018–2024, the REEI coefficient is negative in point-estimate terms but never reaches conventional significance after controls; the negative point estimate flips sign on the exclusion of four distressed banks (Danske Bank, Swedbank, Monte dei Paschi di Siena, Eurobank), and magnitudes are highly sensitive to the choice of CRW weights. We treat Model 2 as a clean null, but we are cautious about its interpretation. The null reflects the joint effect of a short and unbalanced seven-year window (the standardised CQ5 disclosure format applied to only the final three years for most banks), a small cross-section ($N = 32$) that delivers limited power once bank fixed effects absorb the cross-bank variation in the slow-moving REEI, and researcher-calibrated CRW weights. These design and measurement features make a clean test of H2 difficult within the present data; the appropriate reading is that the sectoral composition of European bank lending, as measured by Battiston-derived climate-transition risk weights, is not detectably associated with realised credit deterioration over our window, not that the underlying mechanism is absent. This is consistent with the Bolton et al. (2020) “green swan” framing, in which transition risk is foreseeable but its realisation timing is uncertain, and reinforces the case for longer panels and forward-looking credit risk measures in future work.

Hypothesis 3 (sector-based transition-risk measures provide stronger explanatory power for

bank credit risk than aggregate ESG scores) is not supported by our data. The horse-race specifications in Section 4.6 provide the cleanest empirical articulation of the paper’s two-channel framing. On the identical 149-observation overlap sample, the Bloomberg ESG score is negative and significant at the 5% level ($-1.00, p = 0.043$), the REEI is statistically indistinguishable from zero, and joint inclusion of the two regressors leaves the ESG coefficient essentially unchanged. The within-bank correlation between the two regressors is only 0.27. In direct contrast to H3, the aggregate ESG score, not the REEI, carries the detectable credit-risk-relevant signal on our sample. The fact that the two regressors generate largely independent variation also means the conclusion is not a collinearity artefact: ESG ratings and sectoral transition-risk exposure capture different empirical information about a bank in our data, even though the conceptual distinction does not require them to be fully independent in every setting: a bank’s lending structure may itself influence how external raters score its ESG performance. The two-channel finding should be read in light of the construction asymmetry between the regressors (Section 5.4) and on the understanding that a longer panel and a refined REEI could in principle reverse the comparison.

A natural reading of these three results jointly is that the credit-risk-relevant content of aggregate ESG scores, on our window, is largely social and governance rather than environmental: the Social pillar drives H1, while a transition-risk measure constructed from the sectoral composition of the loan book (the REEI) does not deliver an additional, separate signal. We frame this as a hypothesis about why our pattern looks the way it does (the channel through which ESG associates with NPL ratios in 2018–2024 European banking is not primarily a climate-transition channel) rather than a conclusion. It cannot be tested directly within the present specifications, but it follows naturally from the pillar pattern and is consistent with the difficulty of detecting a transition-risk signal in realised NPL data over a window in which most of the realised credit cycle is dominated by post-GFC deleveraging.

The paper’s contribution to the literature is fourfold. *First*, by focusing on banks rather than non-financial corporates, it extends an ESG–credit risk literature that has been dominated by firm-level studies (Hauch, 2026; Apergis et al., 2022) into a setting (bank lending) in which the ESG–credit-risk link operates differently because banks are simultaneously the rated entity and the intermediary whose lending decisions shape the ESG profile of the broader economy. *Second*, it constructs and applies the REEI, a hand-built composite measure of climate-transition exposure built from the Battiston classification (Battiston et al., 2017) and applied to hand-collected Pillar 3 CQ5 disclosures; the index itself is a methodological contribution that, on longer panels with thicker coverage, can be deployed as a regulatory and supervisory tool independent of any specific finding in the present paper. *Third*, the two-model structure empirically separates the perception channel (ESG ratings) from the portfolio-composition channel (sectoral transition-risk exposure) on the same data, allowing the relative importance of the two channels to be tested directly rather than asserted. The horse-race specification (Section 4.6) operationalises this separation and produces, to our knowledge, the first explicit within-bank comparison of the two channels in a European panel. *Fourth*, the pillar decomposition documents that the credit-risk-relevant content of the Bloomberg aggregate ESG score for banks is concentrated in the Social pillar rather than the Environmental pillar, qualifying earlier readings of the ESG–credit-risk link that treat the aggregate score as a homogeneous risk signal (Liu & Xie, 2024; Al-Qudah et al.,

2022) and complementing the rating-divergence and measurement-error concerns documented by Berg et al. (2022, 2024) and Kotsantonis & Serafeim (2019). Taken together, these contributions sharpen the way subsequent work can ask, and answer, the ESG–bank-credit-risk question: not as a single regression on a single composite score, but as a decomposition across pillars and a comparison of perception-side and portfolio-side channels.

The findings have practical implications for bank management, supervisors, and the design of sustainable finance regulation. For bank management, the consistent negative direction of the ESG coefficient across specifications, robustness checks, and pillar decompositions provides suggestive evidence that ESG-conscious banks tend to experience lower credit deterioration, with the Social dimension carrying most of the signal in our data. For supervisors, the finding that the ESG score and the REEI capture largely independent empirical variation means that a bank’s aggregate ESG score should not be used as a proxy for the climate alignment of its lending book; granular portfolio-level analysis of the kind enabled by the EBA Pillar 3 ESG ITS is a complement to, not a substitute for, ESG rating-based assessments. In a financial system increasingly focused on climate transition risk, understanding whether ESG labels or portfolio fundamentals better predict credit outcomes is not merely an academic question; it is a question with direct implications for bank supervision, credit risk management, and the design of sustainable finance regulation. The hand-collected Pillar 3 data, while limited in window length, demonstrate the value of granular sectoral analysis that goes beyond aggregate ESG scores, and the two-model framework can serve as a template for more powerful tests of the ESG–credit risk nexus in European banking as regulatory disclosure requirements continue to improve and time series lengthen.

References

- Aburto Barrera, L. I., & Wagner, J. (2023). A systematic literature review on sustainability issues along the value chain in insurance companies and pension funds. *European Actuarial Journal*, 13, 653–701.
- Al-Qudah, A. A., Hamdan, A., Al-Okaily, M., & Alhaddad, L. (2022). The impact of green lending on credit risk: evidence from UAE’s banks. *Environmental Science and Pollution Research*, 30(22), 61381–61393.
- Apergis, N., Poufinas, T., & Antonopoulos, A. (2022). ESG scores and cost of debt. *Energy Economics*, 112, 106186.
- Battiston, S., Mandel, A., Monasterolo, I., Schütze, F., & Visentin, G. (2017). A climate stress-test of the financial system. *Nature Climate Change*, 7(4), 283–288.
- Berg, F., Kölbel, J. F., & Rigobon, R. (2022). Aggregate confusion: The divergence of ESG ratings. *Review of Finance*, 26(6), 1315–1344.
- Berg, F., Kölbel, J. F., Pavlova, A., & Rigobon, R. (2024). ESG confusion and stock returns: Tackling the problem of noise. *Working Paper*, MIT Sloan / NBER.
- Bloomberg L.P. (2025). *Bloomberg ESG Scores: Methodology*. Bloomberg Professional Services. Available at: <https://professional.bloomberg.com/globalassets/professional/solutions/sustainable-finance/scores/bloomberg-esg-scores-methodology.pdf>
- Bolton, P., Després, M., Pereira da Silva, L. A., Samama, F., & Svartzman, R. (2020). *The green swan: Central banking and financial stability in the age of climate change*. Bank for International Settlements.
- Cameron, A. C., & Miller, D. L. (2015). A practitioner’s guide to cluster-robust inference. *Journal of Human Resources*, 50(2), 317–372.
- Carnevale, C., & Drago, D. (2024). Do banks price ESG risks? A critical review of empirical research. *Research in International Business and Finance*, 69, 102227.
- European Central Bank (2022). *2022 Climate Risk Stress Test*. ECB Banking Supervision Report, July 2022.
- Freeman, R. E. (1984). *Strategic management: A stakeholder approach*. Pitman.
- Friede, G., Busch, T., & Bassen, A. (2015). ESG and financial performance: Aggregated evidence from more than 2000 empirical studies. *Journal of Sustainable Finance & Investment*, 5(4), 210–233.
- Friedman, M. (1970). The social responsibility of business is to increase its profits. *The New York Times Magazine*, September 13.
- Hauch, S. (2026). ESG rating divergence and corporate credit risk. *The Journal of Risk Finance*, 27(1), 132–157.

- Hoepner, A., Oikonomou, I., Scholtens, B., & Schröder, M. (2016). The effects of corporate and country sustainability characteristics on the cost of debt: an international investigation. *Journal of Business Finance & Accounting*, 43(1–2), 158–190.
- Hong, H., & Kacperczyk, M. (2009). The price of sin: The effects of social norms on markets. *Journal of Financial Economics*, 93(1), 15–36.
- Kotsantonis, S., & Serafeim, G. (2019). Four things no one will tell you about ESG data. *Journal of Applied Corporate Finance*, 31(2), 50–58.
- Liu, Y., & Xie, Z. (2024). ESG performance and bank risk: Evidence from Chinese listed banks. *Finance Research Letters*, 60, 104878.
- van Duuren, E., Plantinga, A., & Scholtens, B. (2016). ESG integration and the investment management process: Fundamental investing reinvented. *Journal of Business Ethics*, 138(3), 525–533.

A. Pillar 3 Data Extraction Notes

This appendix documents the data extraction process for each bank’s Pillar 3 sectoral loan disclosure, including format breaks, scope differences, and reconciliation procedures. For each bank, we note: (a) the years for which NACE-level data are available in a consistent format, (b) the years for which alternative taxonomies or broader scope definitions apply, (c) the specific variable definitions used (“non-performing” vs. “defaulted” vs. “Stage 3”), and (d) any sum-vs-disclosed-total discrepancies.

Coverage by year

The Model 2 sample comprises 151 bank-year observations across 32 banks over 2018–2024 (after dropping one bank-year with `total_gross = 0`, namely Monte dei Paschi 2018). Coverage is sparse in early years and dense from 2021 onward:

- 2018: 2 banks reporting (Eurobank and Monte dei Paschi). Monte dei Paschi’s observation has `total_gross = 0` and is dropped, leaving 1 valid bank-year (Eurobank) in 2018.
- 2019: 10 banks reporting, predominantly the early-adopter banks with EBA-format Pillar 3 templates (Italian and French banks lead).
- 2020: 16 banks reporting; coverage broadens as more banks adopt NACE-format sector breakdowns.
- 2021: 28 banks reporting.
- 2022–2024: 32 banks reporting in each year, reflecting the binding effect of the EBA’s ITS on Pillar 3 ESG disclosures.

Format heterogeneity

Banks reporting in 19-division NACE Rev2 format throughout the available window: BNP Paribas, BBVA, Banco Santander, CaixaBank, Crédit Agricole, Société Générale, ING Group, Intesa Sanpaolo, UniCredit, Banco BPM, Erste Group, KBC Group. For these banks the mapping to our six-bucket Battiston classification is direct. KBC reports REEI in percent format whereas other banks use decimal format; the loader (cell 18 of `Regression.ipynb`) detects and harmonises both representations to a uniform decimal scale.

Banks requiring manual mapping from broader/alternative taxonomies: Deutsche Bank and Commerzbank report sector breakdowns in their pre-2021 Pillar 3 reports using broader, bank-specific aggregations that do not correspond directly to NACE Rev2 divisions. We map these to NACE letters where the underlying sub-sector descriptors permit unambiguous assignment, and exclude observations where the mapping would require speculative imputation.

Banks with shorter coverage (3–4 years available): ABN AMRO (4 years, IPO-related history limits earlier coverage), Barclays, HSBC, Lloyds Banking Group, NatWest Group (UK banks, with first NACE-format reporting from 2022 in most cases), Danske Bank (3 years).

Definition inconsistencies

CaixaBank reports “defaulted” exposures at the sector level rather than “non-performing” exposures. Under CRR definitions, “default” and “non-performing” are closely related but the NPE definition is slightly broader. We treat CaixaBank’s “defaulted” figures as a close proxy for NPL exposure. Since the regression input from the Pillar 3 data is the gross carrying amount per sector (not the NPL amount), this discrepancy does not affect the REEI calculation; it would affect a sector-level NPL-rate analysis that we do not pursue here.

Stage 3 vs. NPE mapping. For banks reporting under IFRS 9 stage classification (Stage 3), we treat Stage 3 exposures as conceptually equivalent to NPE for the purpose of identifying impaired exposures. The two concepts differ in detail (Stage 3 is forward-looking under the IFRS 9 expected-loss framework, NPE is backward-looking under the EBA harmonised definition), but both target the same underlying portfolio of impaired loans.

Sum-vs-total reconciliation

For each bank-year, we compare the sum of sector-level gross carrying amounts to the bank’s disclosed total gross loans. The `check_shares` variable in the consolidated REEI panel reports this reconciliation ratio. The vast majority of bank-years reconcile within 0.5% of unity. The single exception (Monte dei Paschi 2018, `check_shares` = 0) is dropped from the sample because `total_gross` = 0 indicates a Pillar 3 disclosure gap rather than a true zero-loan-book observation.

Currency conversion

All Pillar 3 figures are reported in the bank’s reporting currency (EUR, GBP, CHF, SEK, NOK, DKK, or PLN). Since the REEI is a unit-free ratio (sector share weighted by CRW), no currency conversion is required for the REEI itself. The `total_gross` column is reported in the original currency for transparency but is not used directly in the regression.